# DBDesigner Fork

Manual · Version 2.1.2

Visual database design for MySQL, SQLite and Firebird
https://github.com/Teimue/dbdesigner-fork

DBDesigner 4 © 2003 fabFORCE.net · DBDesigner Fork © 2006-2026 the contributors · GNU GPL

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# General Information

DBDesigner Fork is a free database design system that integrates database design, modelling, creation and maintenance into a single, seamless environment.

It is a community fork of DBDesigner 4, written by fabFORCE.net. The fork was started in 2006; since version 2.0 it is built with Free Pascal and Lazarus instead of Delphi and Kylix.

## Platforms and Databases

DBDesigner Fork runs on

- Windows 10 and Windows 11, 64 bit. The program is DPI aware and scales its dialogs, symbols and the model with the display.
- Linux with GTK2 (built from the source code).

The following databases can be connected to, for the reverse engineering, the synchronisation, the Query Mode and the plugins:

- **MySQL 8** (client library `libmysql.dll` / `libmysqlclient.so.21`)
- **SQLite 3** (client library `sqlite3.dll` / `libsqlite3.so.0`)
- **Firebird 3 or newer**, as a server or with the embedded engine (client library `fbclient.dll`)

Oracle, MS SQL Server and ODBC are still listed as drivers in the connection dialogs, but there are no connectors for them yet. The SQL script can be written for more databases than can be connected to (see Export SQL Creates).

## Installation

On Windows the setup program from the releases page installs the program, the plugins, this documentation and the client libraries for SQLite, MySQL and Firebird (with the embedded engine); nothing else has to be installed. The example models and the example databases are copied to `Documents\DBDesigner`. Optionally the setup program installs the MCP server and registers it in Claude Code and Claude Desktop.

The settings, the list of database connections (`DBConn.ini`) and the list of recent files are stored in `%APPDATA%\DBDesigner4\` on Windows and in `~/.DBDesigner4/` on Linux.

The user interface is available in English and German. The language is chosen in [Options]->[DBDesigner Options].

## License

DBDesigner Fork is available for free download and is published under the GNU GPL License, version 2.

Click here to read through the GNU GPL License.

## Support

The source code, the releases and the issue tracker are on GitHub: github.com/Teimue/dbdesigner-fork. Select [Help]->[Visit GitHub Project] to open the page from the program.

## Bug submission

If you have found a bug or got an idea for new functions and features please open an issue at github.com/Teimue/dbdesigner-fork/issues, or select [Help]->[Submit a bug / feature request].

# Fundamentals

The key to successfully using DBDesigner Fork is understanding database fundamentals.

Read through the topics summarized below - these are the features, techniques, and workflows that are fundamental for working with DBDesigner Fork.

## Models and Databases

In DBDesigner Fork, you always work within a model. A model is a visualization of the meta-information stored in a database (e.g. Tables and Indices, Relations, ...). Although it is possible to store initial data for each table directly in the model, it only represents the meta-information, not the data itself.

You can create and maintain as many models as needed, containing a unlimited number of objects. An object can be a database table with columns and indices, a relation between two tables, a note, ...

Models can be designed by placing these objects onto the model's canvas or can be retrieved from existing databases using the reverse engineering function.

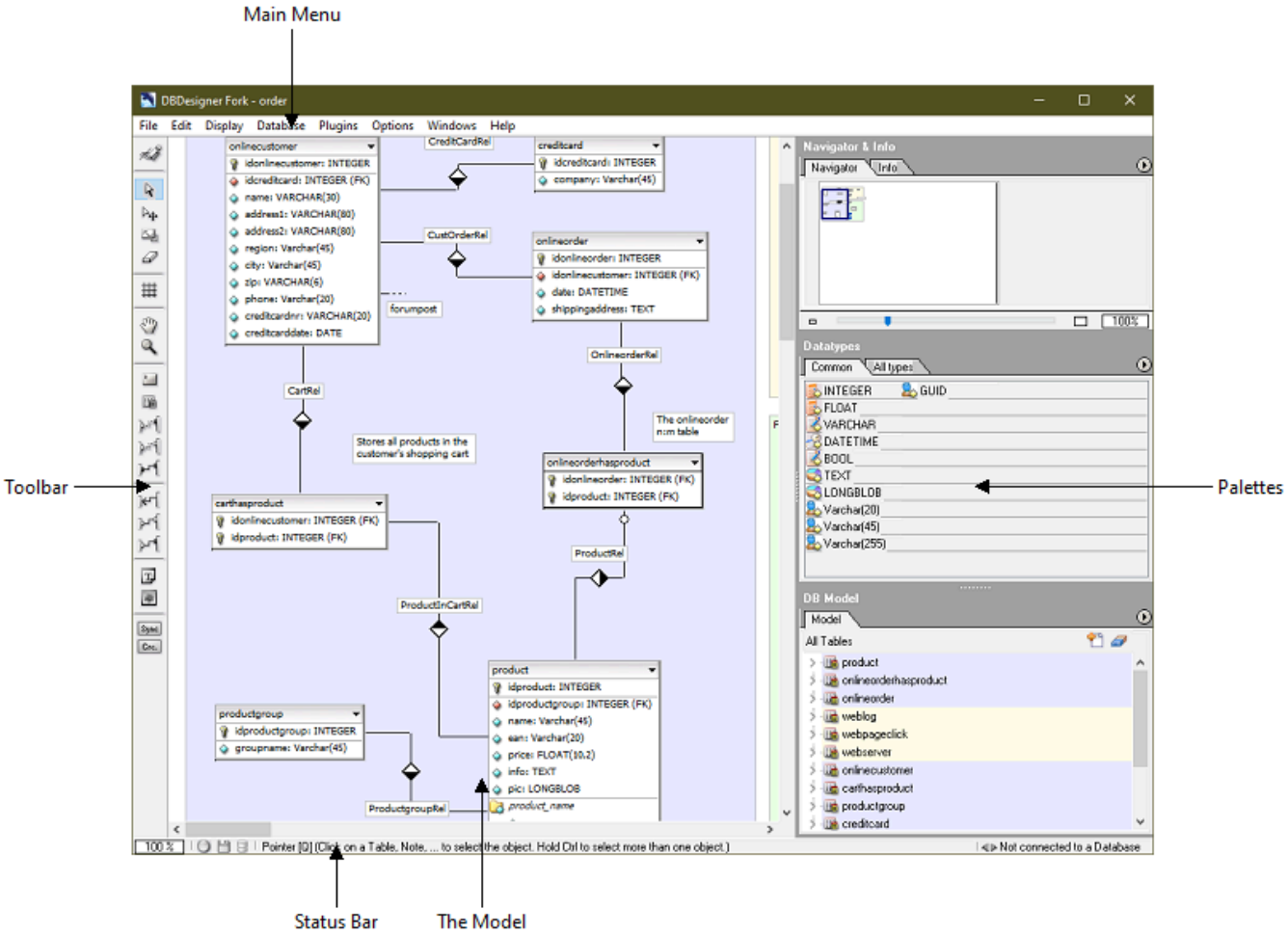
To create the according database the model can be exported as an SQL Creates Script or be created directly from within DBDesigner Fork using the synchronize function. The synchronize function is also used to alter the database automatically when the model has changed.

When DBDesigner Fork is switched to Query Mode the model can be used to build complex SQL querys and edit the tables' data.

The models are saved to XML-files or can be directly stored within the database enabling distributed access to the model.

## The User Interface

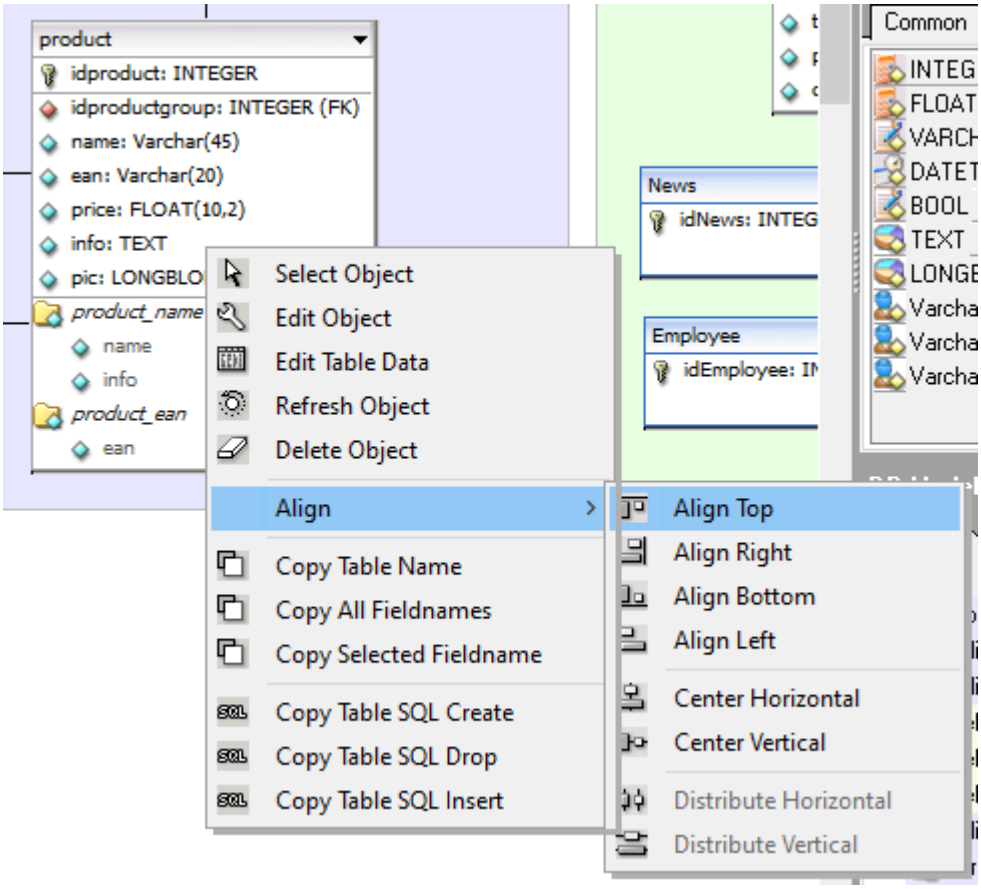
The user interface has been based on industry standard layouting software. This makes creating your database models very easy.




DBDesigner GUI

Beside the familiar window-menus, scrollbars and status-bar DBDesigner Fork provides several palettes (floating windows) which can be used to access frequently used functions more quickly.

DBDesigner Fork makes extensive use of standard windows functions like popup-menus which are accessed by a right mouse click. They provide context sensitive functions for the objects. Drag-n-Drop is used to assigning datatypes to table columns or building table indices.



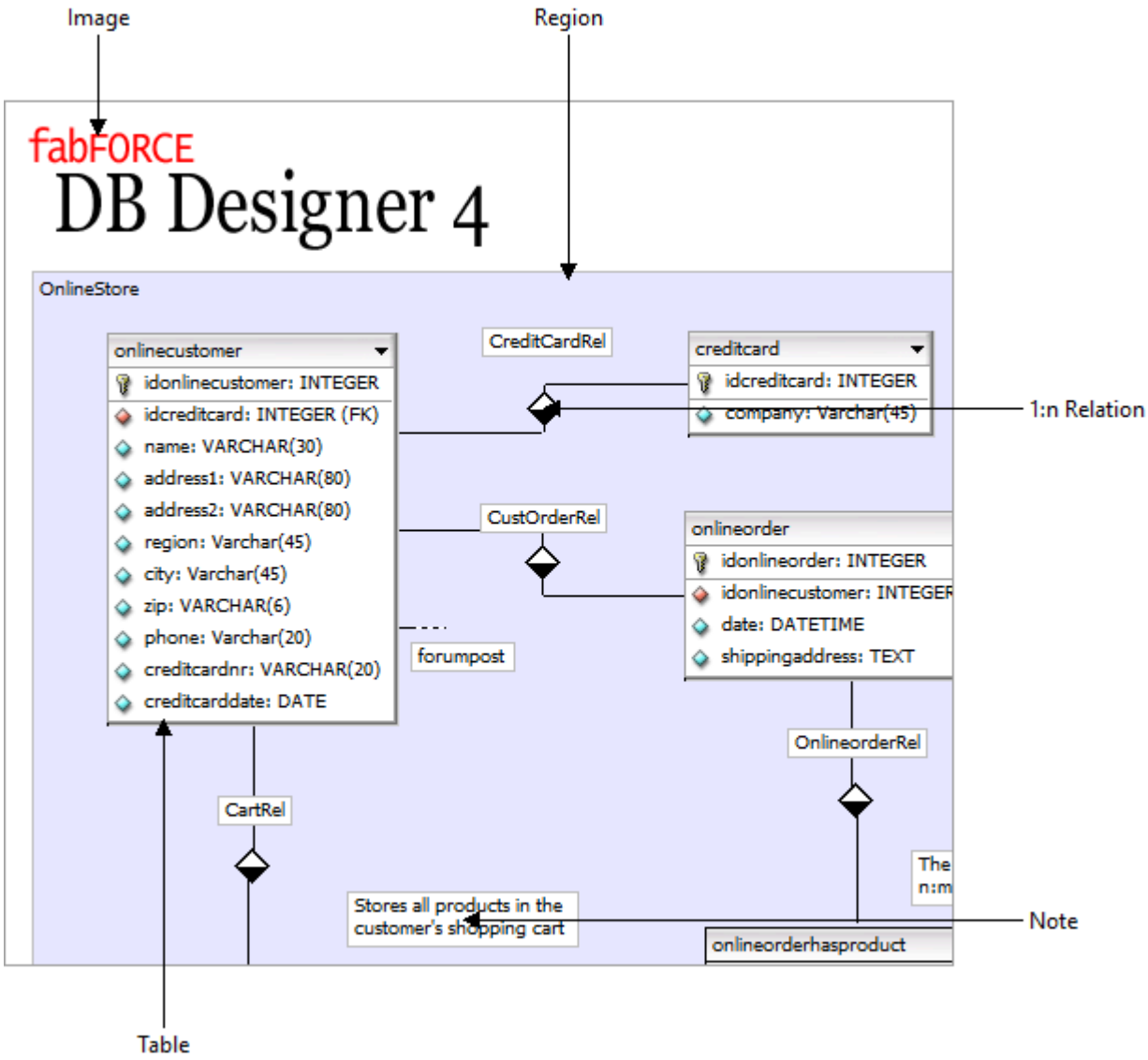

The Table Popup Menu

DBDesigner Fork supports the Multible Document Interface (MDI) which allows you to open an unlimited number of models at the same time. You can switch between the models and use standard copy and paste commands to exchange objects between the models.

# Modelling

The database model is created by placing several objects on the canvas, specify their attributes and relations.

To place an object you have to select the appropriate tool from the Tools-Bar. Beside the most important objects like Tables and Relations (1:1, 1:1 generalization, 1:n, 1:n non identifying, n:m) you can use Notes, Images and Regions which help to "understand" the model's structure.




The Objects

## Tables

Tables represent database-tables. The table is displayed in a window-like manner. The Table's name is displayed in the titlebar, the table columns are printed below and are indicated by an icon. A key icon means this column is in the table's primary key.

### *Primary Key*

Usually one or more columns are defined as the table's Primary Key (PK). These columns must not contain two or more data values which are equal. That makes it possible to clearly identify each record in the table by the Primary Key (e.g. productnumber).

### *Indices*

To make the database find a specific record in the database more quickly, it is possible to define an index on one or more columns. Indices are also used to improve speed when two or more tables are joined together.

## Relations

Relations can only be placed between two tables. They define the relationship between the tables and can create a Foreign Key reference. Tables can be connected by a one-to-one (e.g. person - address), one-to-many (productgroup - product) or many-to-many (e.g. employee - meeting) relation.

## Notes

Notes are simple text boxes containing information about a table or structure. They can be placed anywhere on the model providing information where needed.

## Images

Images can be placed on the model to visualize additional information.

## Regions

Regions provide space for tables with the same attributes. The attributes are set for the region and apply to all tables placed on the region automatically. Furthermore they can be used by plugins to apply specific functions to a group of tables.

# Modelling

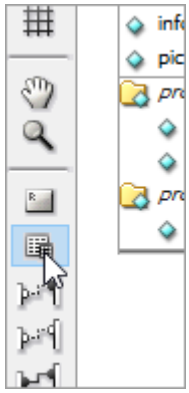
Before starting to model the database the Design Mode has to be selected. To change the current Work Mode click the Work Mode icon in the Tools Palette.

## Tables and Relations

Tables and relations are the core components of each database model. Tables consist of the tablename, the columns and indices and various table options. Relations connect different tables with each other to form one-to-one or one-to-many relations.

### Place a new table

To place a new table onto the model select the Table Tool from the Toolbar or press [T] on the keyboard. The mouse cursow will change to refect the currently selected tool.



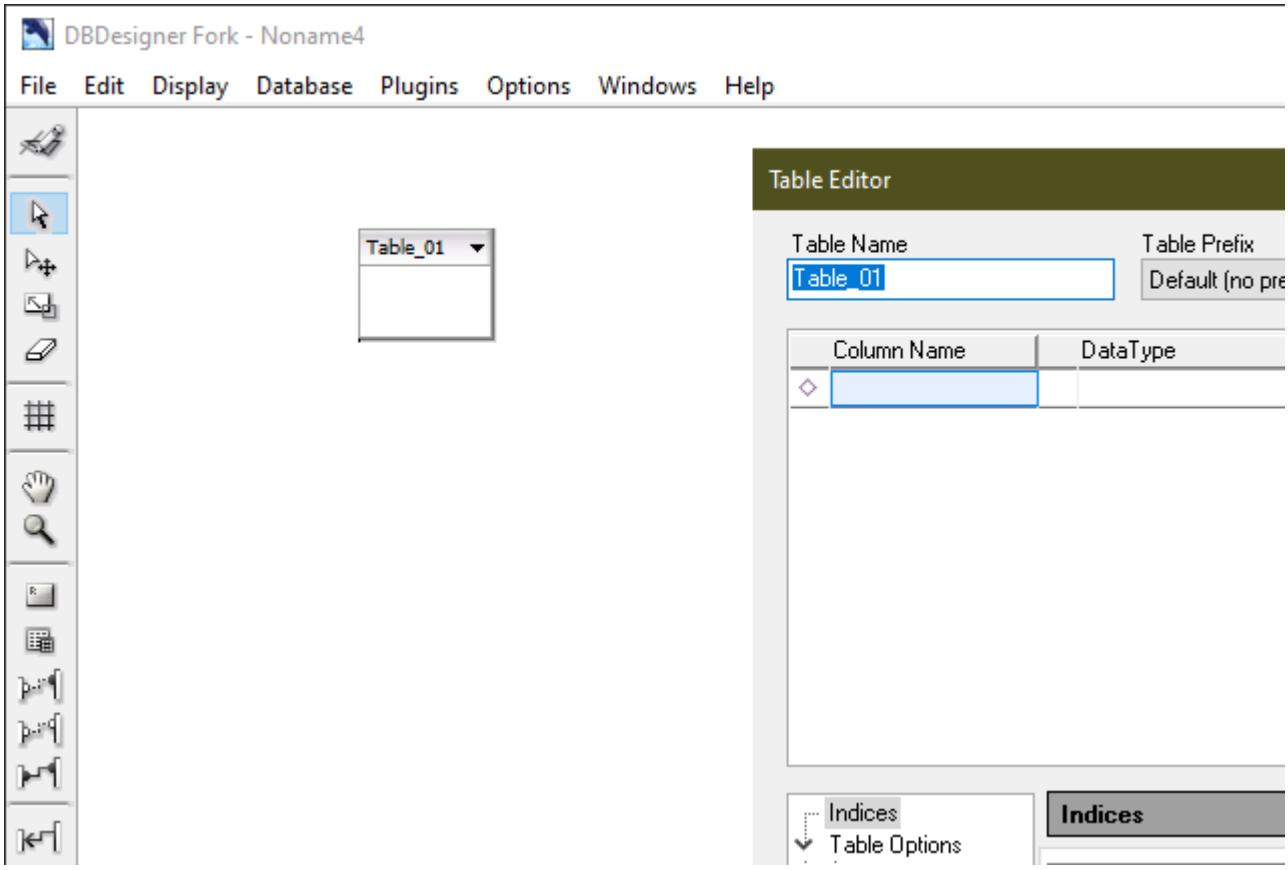
Selecting the Table Tool

To place the table click left on the model. A new table will be created. It will be named [table_XX]. The table's left upper corner will be at the position the mouse was clicked.

After the table was placed the selected tool will change back to the Pointer Tool. The mouse cursor will be changed also.

### Edit a table

To edit an existing table make shure the Pointer Tool is selected. Double click on the table with the left mouse button. The Table Editor will be showen.




Calling the table editor

### *Defining the table's name and columns*

Enter the table's name (e.g. product) and press the return key. The focus will change to the first column. It will be named id+tablename by default. To accept this name press return or overwrite the column's name first.

Now the focus has changed to the second column. Again enter the columns name and repeat for all the table's columns.

Press the Esc key after the last column was named.

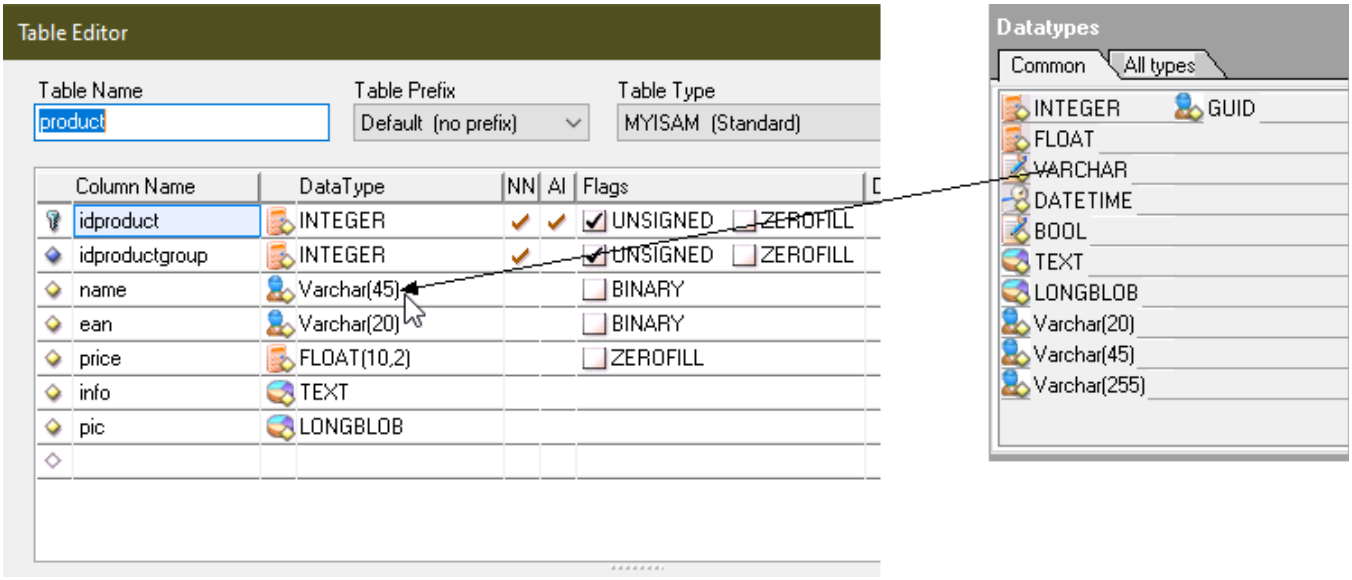
Columns can be reorderd by draq'n'drop. Left click on a column's name, hold the mouse button and drag onto another column. Release the mouse button and the column will be inserted before the column it was dragged on.

### *Assigning the columns' datatypes*

Press the Esc key after the last column was named.

### *Assigning datatypes*

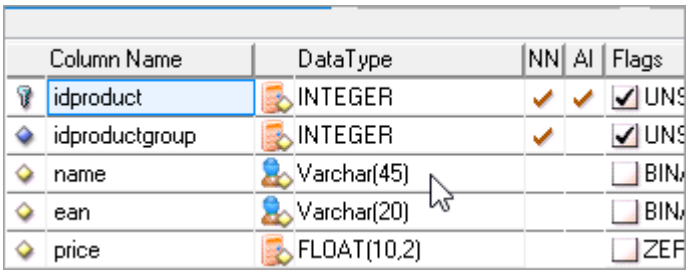
All columns will be created with the default datatype (use Options > Model Options to change the default datatype).




Assigning datatypes

To change the datatype of a column goto the Datatypes Palette and drag the appropriate datatype onto the column in the Table Editor. If the assigned datatype has parametes (e.g. VARCHAR(xxx)) enter the wanted value. Press Return to apply the changes or Esc to discard. Repeat this for all table columns.

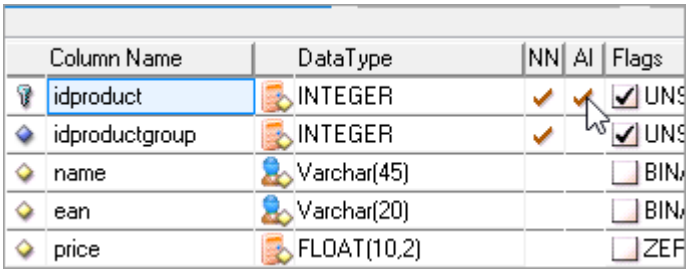
To change a datatypes parameters doubleclick on the datatype and enter the new value.




Setting datatype parameters

## *Primary key, AutoInc and Flags*

The first column has automatically been defined as primary key of the table indicated by a Key Icon left to the columns name. To remove a column from the primary key click the Key Icon. To add a column to the primary key click onto the Column Icon.



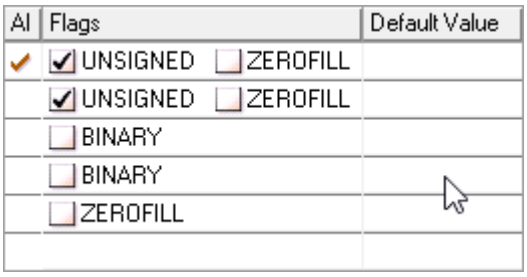

Setting the AutoIncrement property

To make a column automatically incrementing after each insert click the columns row in the AI column.

Each datatype has specific flags. They can be activated and deactivated by clicking.

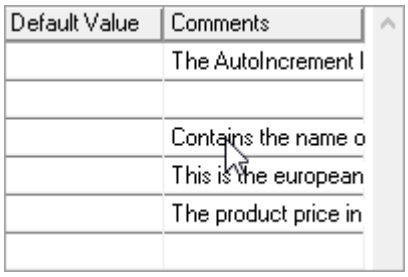
## *Default Values and Comments*

To set the default value of a column doubleclick in the Default Value column and enter the value. Press Return to apply the changes.




Setting Default Values

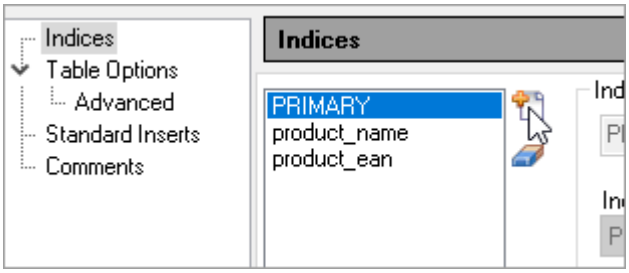
To display the columns' Comments click the icon right to the Default Value column's caption. To edit a comment doubleclick in the Comments column.




Edit the columns' comments

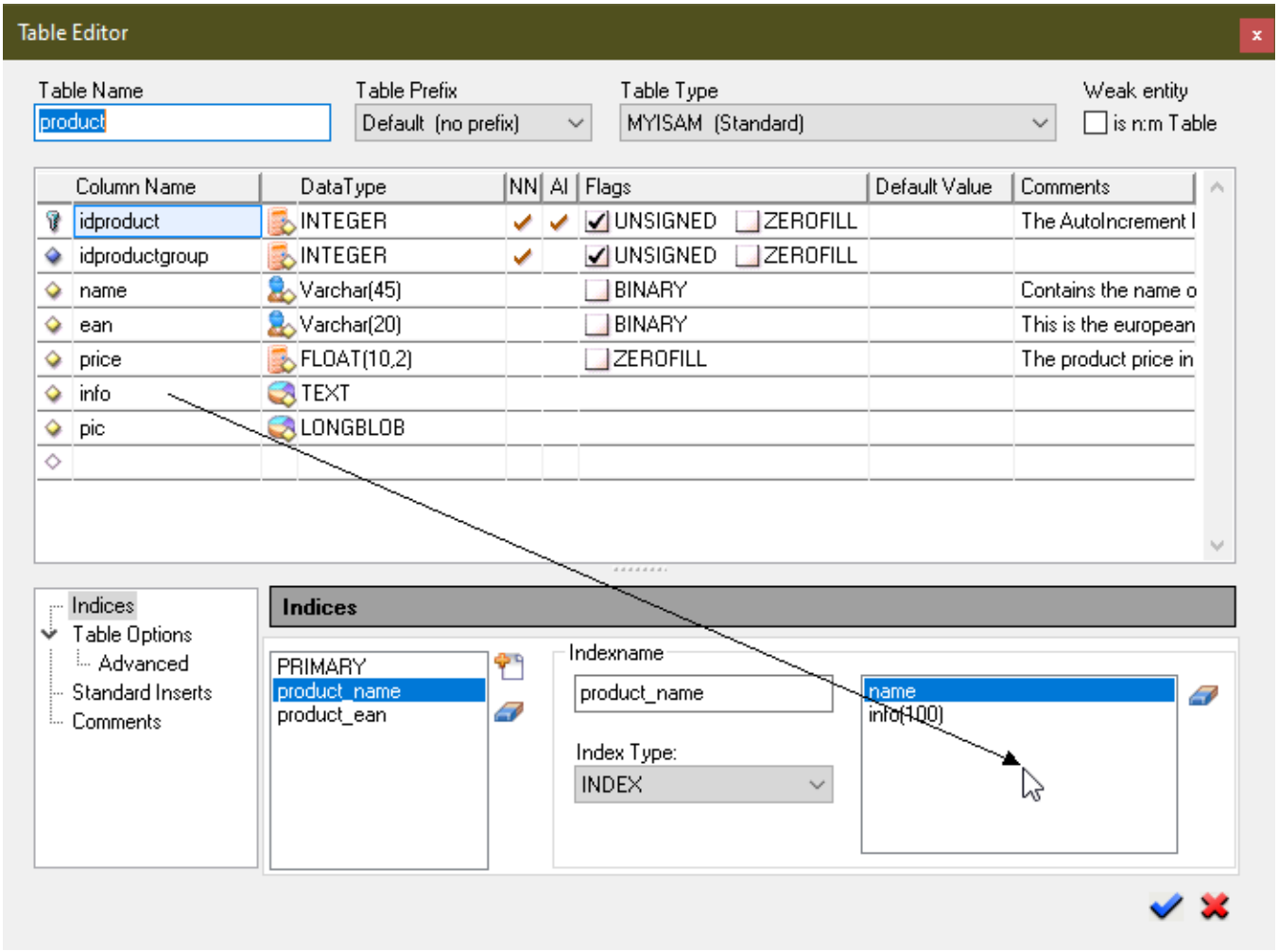
## Create Table Indices

Change to the Indices Tab and click the Plus Icon to add a new index. A String Input dialog will appear. Enter the index's name and press return. The new index is displayed. Now choose the type of the index.




Create a new table index

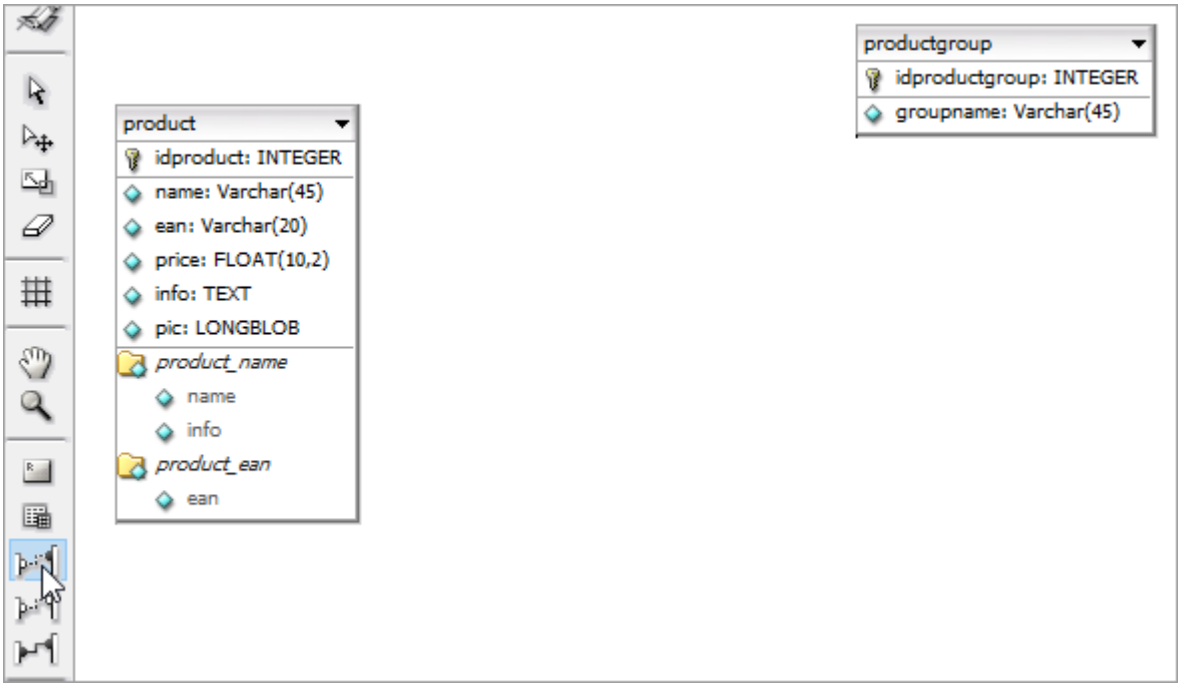
To add a column to the index start dragging the column down to the Columns Listbox and drop it. You can reorder the columns by dragging a column onto another. To delete a column press the Delete button.




Adding columns to the index

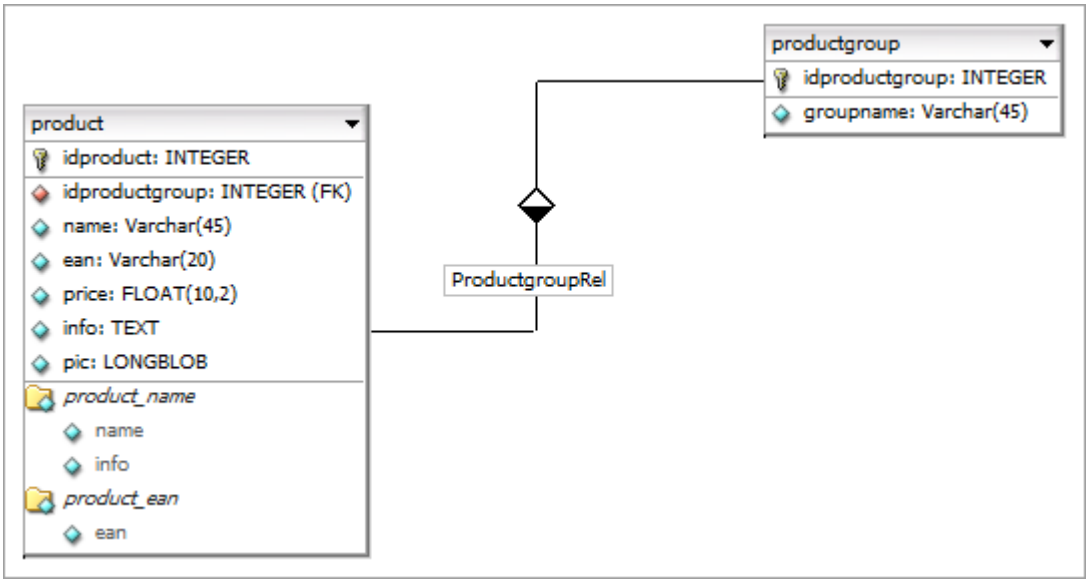
## Make a relation

Add another table to the model using the Table Tool (e.g. productgroup) and define the table's columns. Now select the 1:n Relation Tool (Non-Identifying Relation) from the Toolbar.



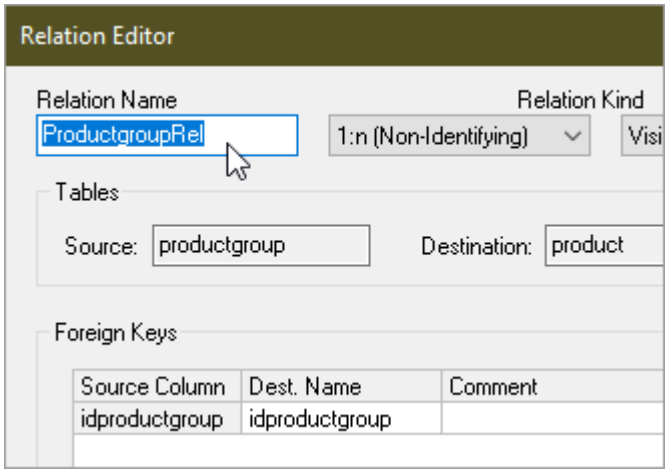

Selecting the 1:n Relation Tool (Non-Identifying Relation)

Click on the first table (productgroup) then on the second table (product) to establish the relation. Note that the second table (product) now has a foreign key column identified by (FK) right to the datatype.




Tables connected by a relation.

Doubleclick on the relation to enter the relation's name. Close the Relation Editor to apply the changes to the model.




Tables connected by a relation.

# Querying

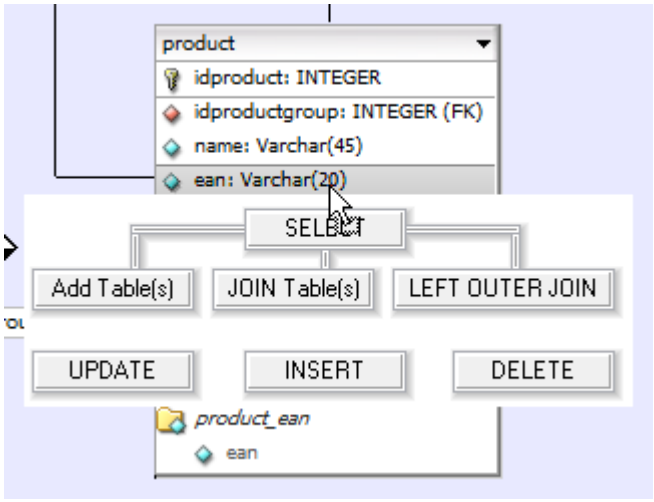
Before starting to query the database the Query Mode has to be selected. To change the current Work Mode click the Work Mode icon in the Toolbar or the Tools Palette or select Query Mode from the Display Main Menu.

## Building selects

SQL selects can be easily build by drag'n'drop commands. Make shure the current worktool is set to a Pointer Tool before starting.

### Building a simple SQL SELECT

To build a SQL select from one table simply click on a table with the left mousebutton and keep holding the button. Now move the mouse cursor down. The Drag Menu is displayed.




The Drag Menu

Keep holding the left mouse button and move the mouse cursor to the SELECT button. Release the mouse button. The SELECT statement will be build and displayed in the SQL Command Editor.

If the table which has to be queried is not visible in the current visible area of the model, click somewhere on the model to set the focus to the model. Now press the [Space] key and drag the model till the table becomes visible.

### Joining tables

To join another table with this table click on the table to join. Start draging like before and release the mouse button over the JOIN Tables(s) Button. The two tables will be joined based on the relations between the tables.

```
SELECT *
FROM productgroup productgroup, product product
WHERE productgroup.idproductgroup=product.idproductgroup
```

Two tables joined

Another possibility is to select all tables that have to be joined by holding the Ctrl Key and clicking on the tables. Then start dragging and release the mouse button over the JOIN Tables(s) button. All tables will be joined based on the relations between them.

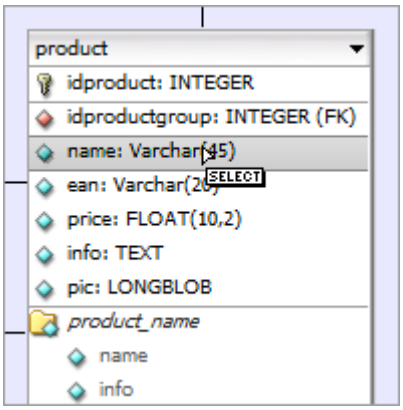
### *Add tables without joining*

To add one or more tables to a query without performing a join drag the table(s) on the Add Table(s) button. They will be added to the SQL command.

## Selecting table columns

When a SQL command is built using the Drag Menu all columns of the table(s) are selected using the SELECT * clause.

To select specific table columns choose the SELECT Clause Tool from the Toolbar or press Ctrl + S. The mouse cursor will change to the SELECT Clause cursor. Now move the mouse over the table and watch the columns be highlighted. Move the mouse cursor to the column you want to select and press the left mouse button.




Selecting a table's column

To column will be added to the SELECT Statement.

### *Add columns to other clauses*

Use the different Clause Tools from the Toolbar to add table columns to the SQL command.

| Tool | Key | Position |
|---|---|---|
| SELECT Clause Tool | Ctrl + S | SELECT [column] FROM ... |
| FROM Clause Tool | Ctrl + F | SELECT * FROM table_a LEFT OUTER JOIN table_b ON [column] ... |
| WHERE Clause Tool | Ctrl + W | SELECT * FROM table_a WHERE [column] ... |
| GROUP Clause Tool | Ctrl + G | SELECT * FROM table_a GROUP BY [column] ... |
| HAVING Clause Tool | Ctrl + H | SELECT * FROM table_a GROUP BY col1 HAVING [column] ... |
| ORDER Clause Tool | Ctrl + O | SELECT * FROM table_a ORDER BY [column] ... |
| SET Clause Tool | Ctrl + E | UPDATE table_a SET [column] ... |
| Pointer Tool | Ctrl + Q | - |

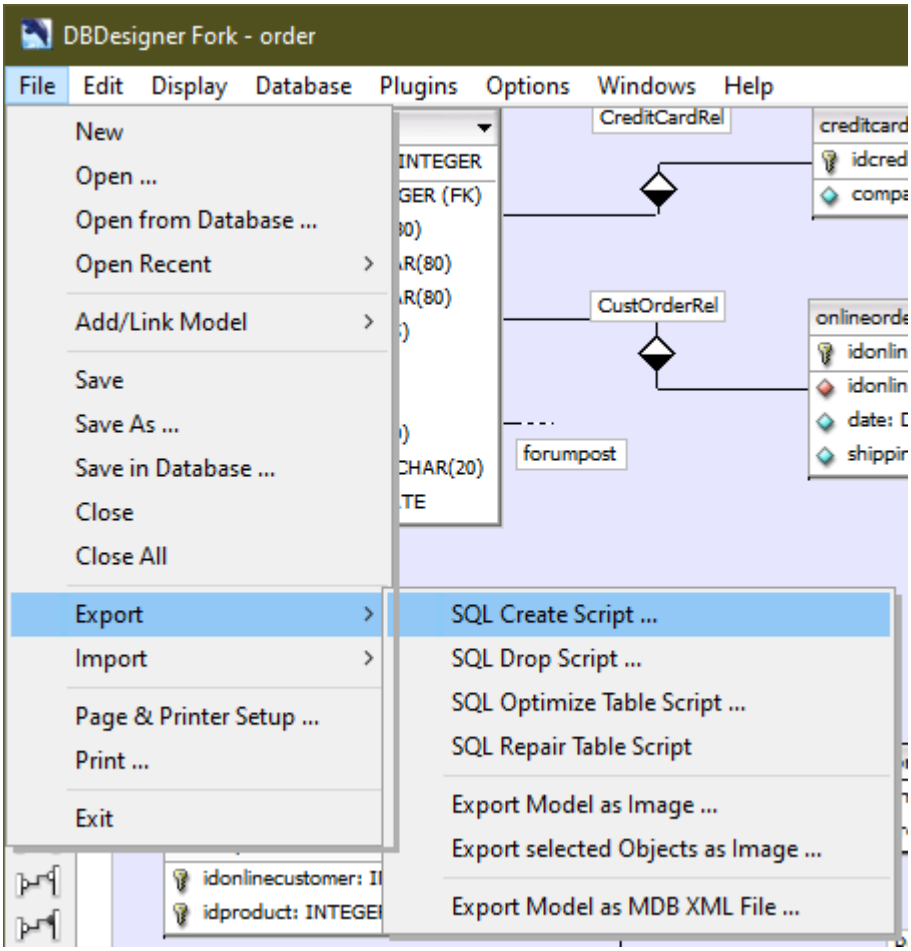
# The User Interface

## Main Elements

### The Main Menu

Through the main menu you gain access to general functions of the program. There are eight main menu items, File, Edit, Display, Database, Plugins, Options, Windows and Help. The Plugins menu is shown only when plugins are installed.

### File Menu




The File Menu

#### *File - New*

File - New creates a new empty model. If you have opened another model before you can switch between the new and the other model using the Windows menu.

#### *File - Open ...*

File-Open ... opens an existing model from your harddisk or mapped network device. DBDesigner creates a new empty model every time it is started so you can start modelling immediately. If you open a model and no changes have been applied to the empty model it is closed automatically and replaced with the opened model.

#### *File - Open from Database ...*

File - Open from Database ... makes it possible to open a model saved in a database. When selected this menu item the DB-Connection Dialog pops up and you have to select the database containing the desired model.

There can be stored more than one model in the database so you have to select the appropriate model then.

### File - Open recent

File - Open recent displays a list of the last opened models to improve speed opening the most used models. If a new model is opened it is added to the list. If there are more than 10 entries in the list the last entry is deleted automatically.

### File - Add/Link Model

File - Add/Link Model lets you place an existing model into the currently opened model, either from a file (Add/Link Model from File) or from a model stored in a database (Add/Link Model from Database). The Place / Link Model window shows the objects of the other model. Select the objects you want and choose [Place]-> [Add selected Objects] to copy them or [Place]->[Link selected Objects] to link them, then click on the canvas where they should be placed. Linked objects stay connected to the other model and can be updated with Refresh Linked Objects; Show Linked Models lists the models the current model is linked to. The entry Add/Link Model from Online Library is disabled, the online library of fabFORCE.net does not exist anymore.

### File - Save

Use File-File-Save ... to save your model to the harddisk or a mapped network device. When you save a model for the first time, you are promt to locate the destination folder and enter the name of the model. The model is added to the File-Open Recent list automatically.

### File - Save As ...

File - Save As ... can be used to save your current model with a new name. You are promt to locate the destination folder and enter the new name of the model.

### File - Save in Database ...

It is possible to store the model directly in the database you work on. A new database table is used to store all models in this database. This new table does not interfere with the database synconisation and reverse engineering functions.

When selected the File-Save in Database ... menu item the DB-Connection Dialog pops up and you have to select the database which will containing the model. You have to enter the desired name of the model and the model will be stored.

### File - Close

File - Close closes the current active model.

### File - Close All

File - Close All closes all open models. You are asked to save every model that has been changed.

### File - Export-SQL Create Script ...

Use this menu item to export the current active model as a SQL Create script which can be executed against the datbase manually to create the nessesary tables. You can use the database synchronise function instead which will create and update your database automatically.

### File - Export-SQL Drop Script ...

Use this menu item to export the current active model as a SQL Drop script which can be executed against the datbase manually to drop the desired tables. You can use the database synchronise function instead which will create and update your database automatically.

### File - Export-SQL Optimize Table Script ... / SQL Repair Table Script

These menu items write the MySQL statements OPTIMIZE TABLE and REPAIR TABLE for the tables of the model. They are only useful for MySQL databases.

### File - Export-Export Model as Image ...

Select this function to export the whole model as an image file. The image can be saved in the PNG or BMP file format. The PNG Format is recommended. The picture has the size of the area covered by the objects of the model.

### File - Export-Export selected Objects as Image ...

Like Export Model as Image, but only the selected objects are drawn.

### File - Export-Export Model as MDB XML File ...

Writes the tables, columns, indices and Standard Inserts of the model to an XML schema file (*.schema) in the format of the PEAR MDB database abstraction layer for PHP.

### File - Import-ERwin 4.1 XML Model ...

The import of ERwin® 4.1 XML models is disabled in this version.

### File - Page & Printer Setup ...

Before printing the model you have to select the appropriate printer and set the page size and orientation as well as the scale factor of the model which enables you to print the model distributed on several pages.

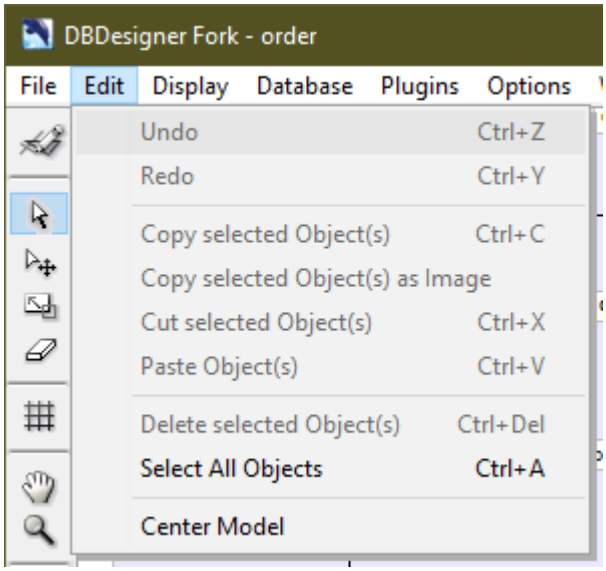
### File - Print ...

File - Print ... lets you select the pages of the model an the number of copies you want to print.

### File - Exit

File - Exit closes the program.

## Edit Menu




The Edit Menu

### Edit - Undo

Use Edit - Undo to take back the last change applied to the model. You can use this command several times to take back all changes you made.

### Edit - Redo

Edit - Redo applies the last undone action again. This command can also be repeated several times when you have undone several actions.

### Edit - Copy selected

When you have selected on or more objects you can use this command to copy the objects to the clipboard. The objects are stored as text in XML format. You can use any other application to paste the objects.

### Edit - Cut selected

This command is similar to Edit-Copy selected but the objects will also be deleted from the model.

### Edit - Copy selected Object(s) as Image

Copies a picture of the selected objects to the clipboard, so it can be pasted into a word processor or an image editor.

### *Edit - Paste*

Edit - Paste is used to insert previous copied objects from the clipboard. Relation are only inserted if both tables have been copied because when the objects are inserted, new object-identifiers are assigned.

### *Edit - Delete selected*

Edit - Delete selected removes all selected objects from the model. When this happens unintended you can use the undo command.

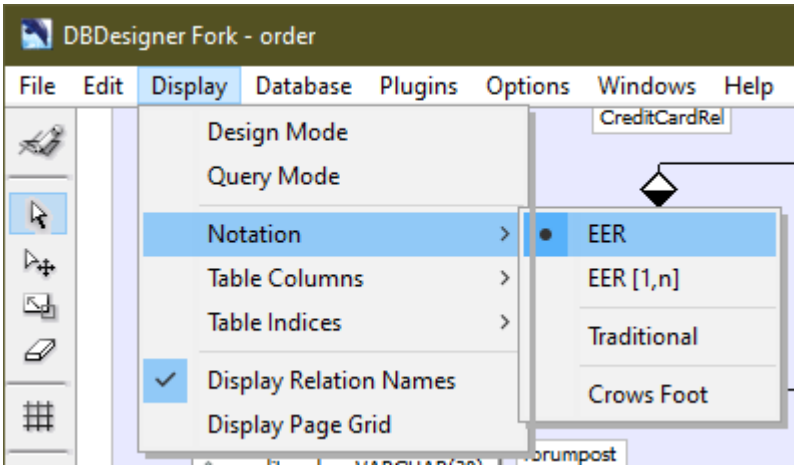
### *Edit - Select All*

Use Edit - Select All to select all objects from the model.

### *Edit - Center Model*

Edit - Center Model moves all objects of the model to the center of the canvas.

## Display Menu



The Display Menu

### *Display - Design Mode*

Select this menu item to change to Design Mode. A checkmark will be drawn left to the menu item when Design Mode is selected.

### *Display - Query Mode*

Select this menu item to change to Query Mode. A checkmark will be drawn left to the menu item when Query Mode is selected.

### *Display - Notation*

Display - Notation specifies the notation used to display the model. When a notation is selected a checkmark will be drawn left to the menu item.

### *Display - Notation - EER*

Display - Notation - EER will set the notation to the common EER notation.

### *Display - Notation - EER [1,n]*

Display - Notation - EER [1,n] will set the notation to another EER notation, displaying intervals.

### *Display - Notation - Traditional*

Display - Notation - Traditional will change to a notation which will be familiar to ERwin® users.

### *Display - Notation - Crows Foot*

Display - Notation - Crows Foot draws the relations in the Crow's Foot notation.

### *Display - Table Columns-Entity Level*

Display - Table Columns will specify how tables are displayed on the model. Use Display - Table Columns-Entity Level to hide all column information.

### *Display - Table Columns-Primary Key Level*

When Display - Table Columns-Primary Key Level is selected, only primary key columns are displayed.

### *Display - Table Columns-Attribute Level*

Use Display - Table Columns-Attribute Level to display all columns of the tables.

### *Display - Table Columns-Physical Schema Level*

Select Display - Table Columns-Physical Schema Level to display the datatypes of the columns.

### *Display - Table Columns-Show Foreign Keys*

If Display-Table Columns-Show Foreign Keys is select foreign keys are shown as columns in the appropriate tables.

### *Display - Table Indices*

This menu item is used to display each table's indices on the model.

### *Display - List Table Indices*

A checkmark will be drawn left to the menu item when table indices are displayed.

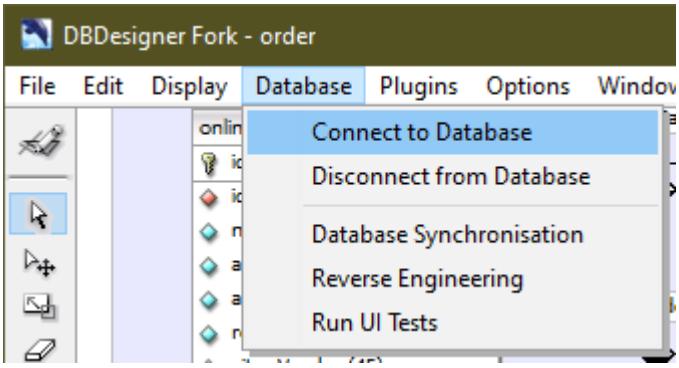
### *Display - Display Relation Names*

Display - Display Relation Names can be selected to display the names of relations.

### *Display - Display Page Grid*

Select Display - Display Page Grid to show the partitioning of printable pages. You can use the Page & Printer Setup Dialog to change the initial page size.

## Database Menu




The Database Menu

### *Database - Connect to Database*

Opens the Database Connection Dialog and connects to a database. The connection is used in Query Mode and by the plugins.

### *Database - Disconnect from Database*

Closes the current database connection. The menu item is enabled only while a connection is open.

### *Database - Database Synchronisation*

Use this function to synchronise your model with a MySQL, SQLite or Firebird database. This function can also be used with an empty database to create all the tables.

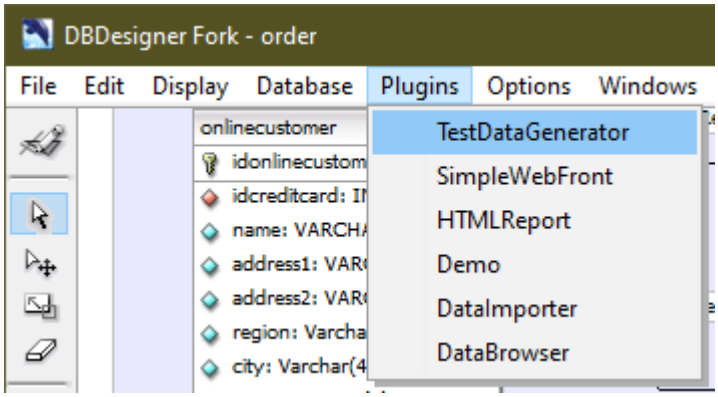
### *Database - Reverse Engineering*

Use this function to create a model based on an existing database.

### *Database - Run UI Tests*

Runs the automated self-test of the user interface: menu items and buttons are clicked one after another and errors are reported. It is meant for developers; it never opens a database connection and leaves your settings untouched, but it creates and closes models, so save your work first.

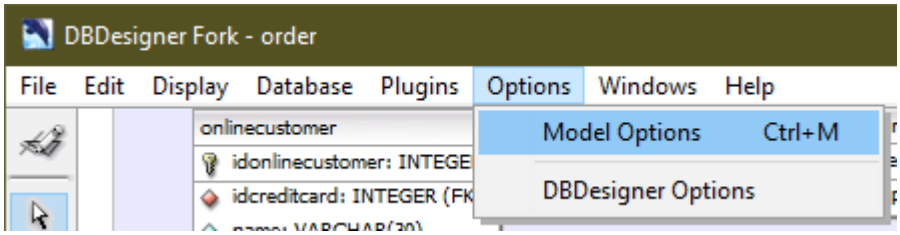
## Plugins Menu




The Plugins Menu

The Plugins Menu lists the DBDesigner Fork plugins found in the program's directory (the programs named DBDplugin_*). It is shown only when at least one plugin is installed. The plugins are described on the page Plugins.

## Options Menu




The Options Menu

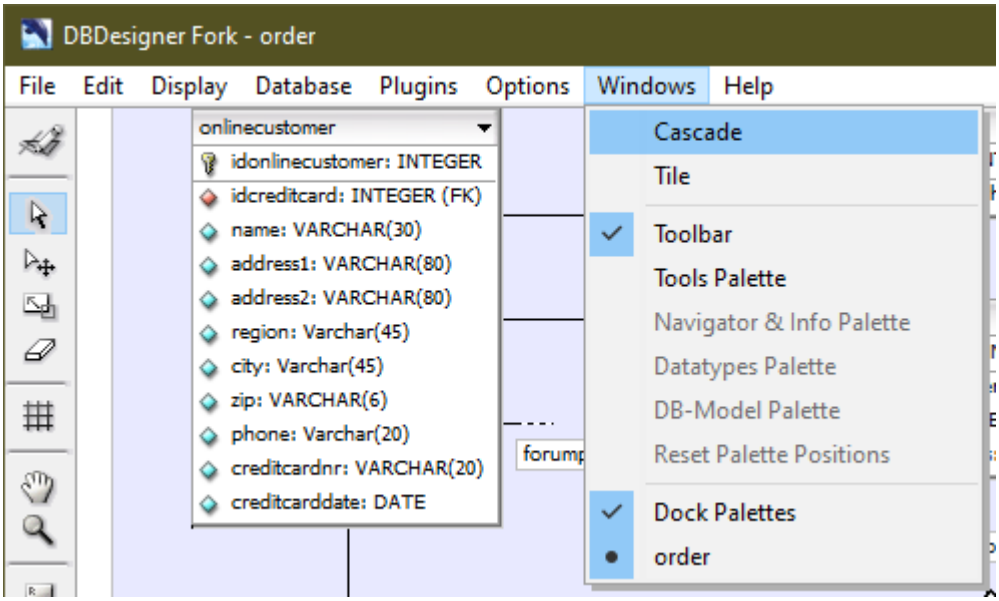
### *Options - Model Options*

Options - Model Options will call the Model Options Dialog where several options can be set for the model like the default datatype or the display font.

### *Options - DBDesigner Options*

Options - Model Options will call the Model DBDesigner Options Dialog where several general options can be set.

## Windows Menu



The Windows Menu

### *Windows - Cascade*

Use this command if you have opened several models and want to arrange the windows in a overlapping style.

### *Windows - Tile*

Use this command if you have opened several models and want to arrange the windows beside each other.

The window styles of DBDesigner 4 (Standard, Motif, SGI, Platinum) do not exist anymore; the program uses the look of the operating system.

### *Windows - Toolbar*

Checked this menu item to display the Toolbar which is docked to the left of the main window.

### *Windows - Tools Palette (MS Windows only)*

If this menu item is checked the Tool-Palette becomes visible. It can be used instead of the Toolbar.

### *Windows - Navigator & Info Palette*

If this menu item is checked the Navigator & Info-Palette becomes visible. If it is unchecked the Navigator & Info-Palette is hidden.

### *Windows - Datatypes Palette*

If this menu item is checked the Datatypes-Palette becomes visible. If it is unchecked the Datatypes - Palette is hidden.

### *Windows - DB-Model Palette*

If this menu item is checked the DB-Model-Palette becomes visible. If it is unchecked the DB-Model-Palette is hidden.

### *Windows - Reset Palette Positions*

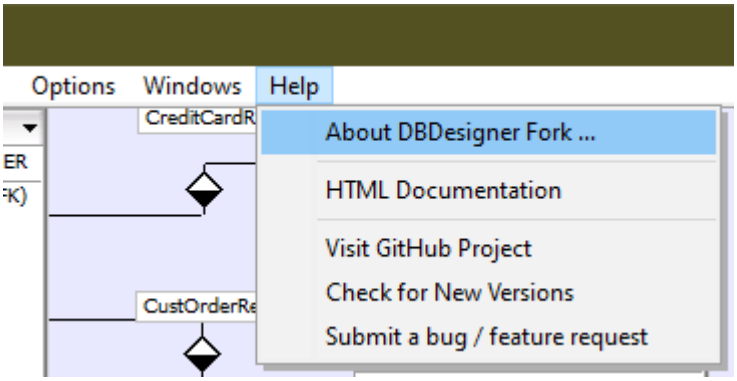
Call this command to reset the palette positions to their initial locations.

### *Windows - Dock Palettes*

Use this menu item to dock or undock the palettes to the right of the main window.

Below this menu item a list of all opened models is displayed. Select a menu item to bring the appropriate menu to the front.

## Help Menu



The Help Menu

### *Help - About DBDesigner Fork ...*

Displays the splash screen with the version of the program.

### *Help - HTML Documentation*

Opens this documentation in the web browser. The browser can be chosen in the DBDesigner Options (Use other HTML Browser). Press F1 in a dialog to open the page about this dialog.

### *Help - Visit GitHub Project*

Opens the page of the project, https://github.com/Teimue/dbdesigner-fork, in the web browser.

### *Help - Check for New Versions*

Asks GitHub for the latest release and compares it with the installed version. If a newer version is available, you are asked whether the download page should be opened. If GitHub cannot be reached, you can open the releases page instead.

### *Help - Submit a bug / feature request*

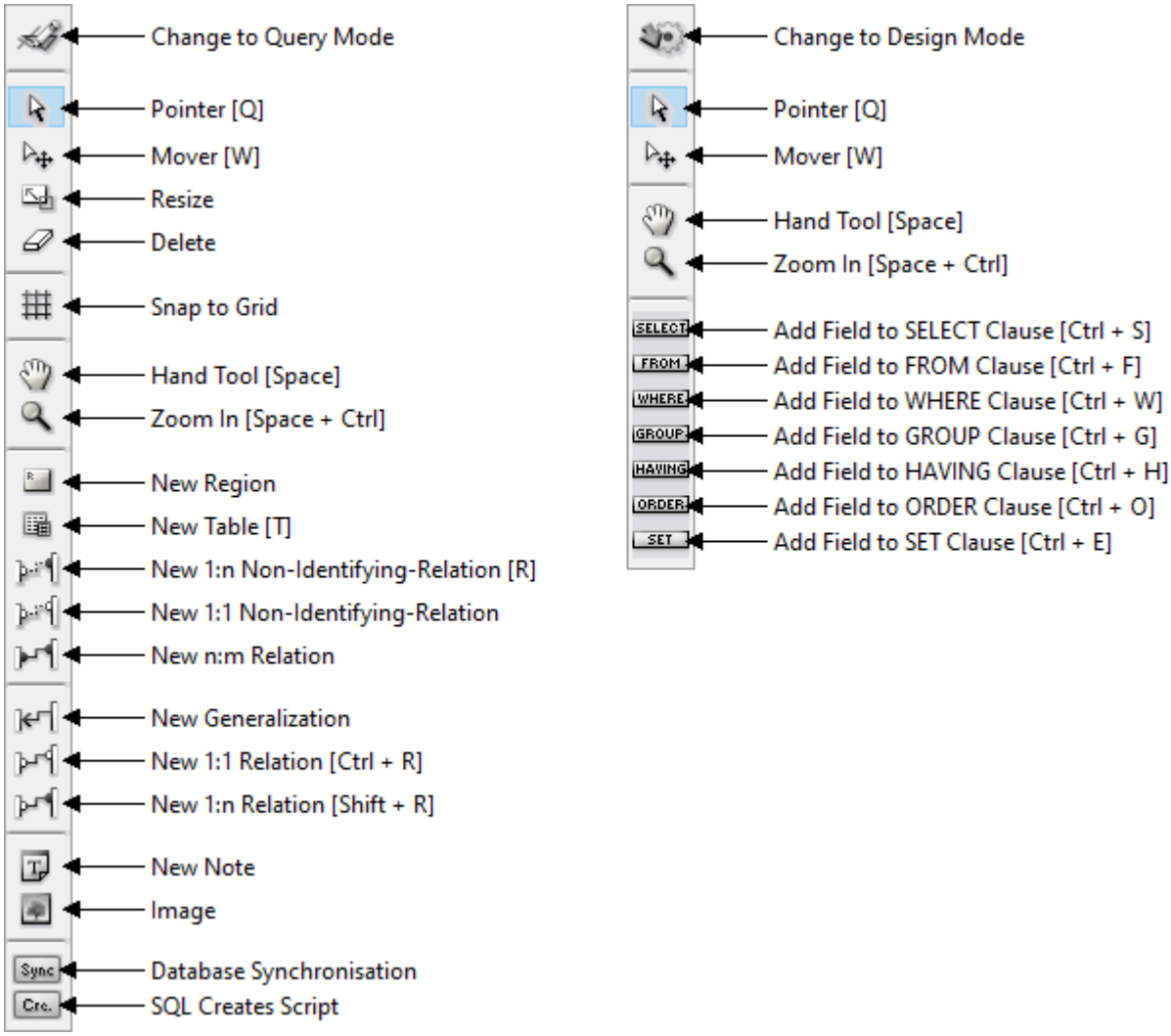
Opens the issue tracker of the project, https://github.com/Teimue/dbdesigner-fork/issues, in the web browser.

# Palettes

Palettes are floating windows which always stay on top and provide fast access to important functions.

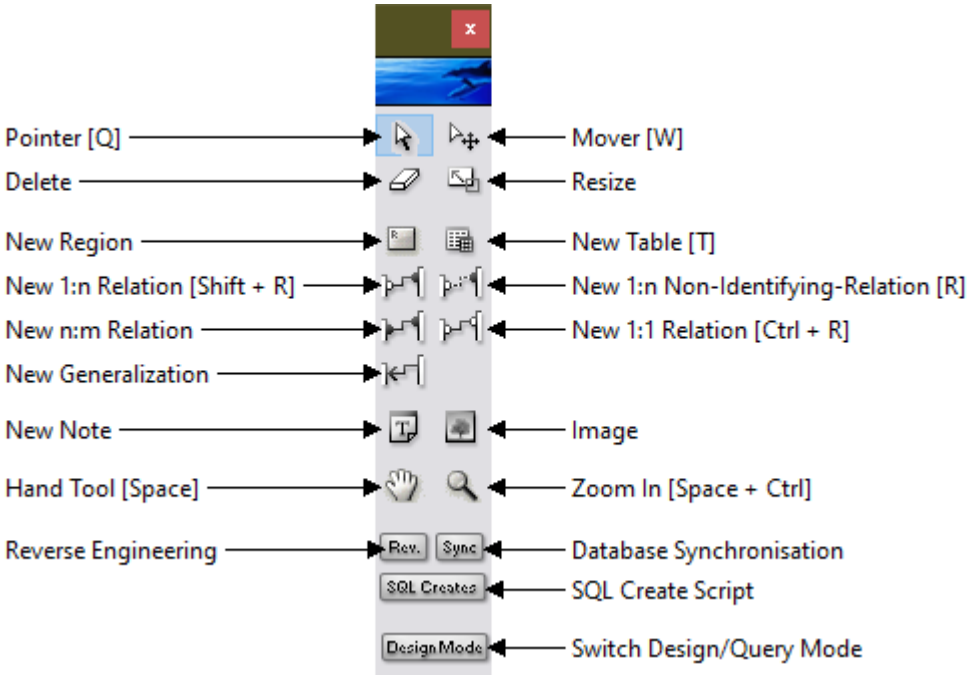
## The Toolbar / Tools Palette

The Toolbar (and the Tools Palette) provides quick access to the most important work tools. (On a Linux system the Tools Palette is disable because of a limitation in the window system.)




The Toolbar

To view the Tools Palette select select [Windows]->[Tools] from the main menu. Deselect [Windows]->[Tools docked] to hide the docked palette. By default the Tools Palette can be found on the upper left corner of the screen.



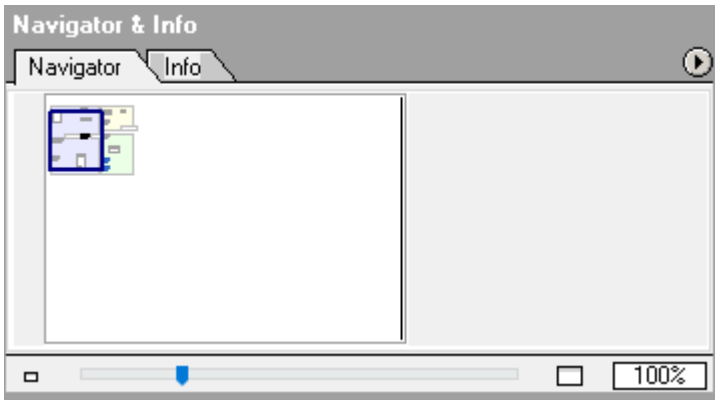

The Tools Palette

Click on the icon to activate the appropriate tool. The mouse cursor will change. Take a look at the Modelling Section to see how to use the single tools.

Clicking on the buttons will call the assigned functions. Use the mode button to switch between the Design- and Query-Mode. In Design-Mode tables can be moved with the selection tool, too. In Query-Mode you have to use the Move-Tool.

## The Navigator Palette

The Navigator Palette (top right) shows the current visible region of the model and can be used to navigate on the model.




The Navigator Palette

### *Navigator Page*

The current visible region is marked by a blue rectangle. When the model is scrolled or moved with the panning tool the marker is moved to the new position automatically. Click on the white area to move the marker and scroll the model to the appropriate position. Depending on the speed of your machine and the size of the model you can also drag the marker.

To change to Zoom-Factor you can use the trackbar or click on the small and large table symbol left and right to the trackbar. To enter the zoom factor manually click on the zoom edit, type in the desired value (without percent charater) and press enter.

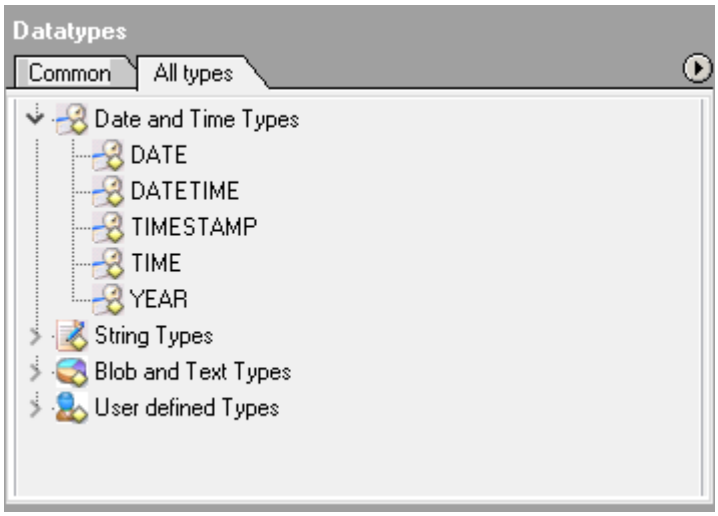
Click on the upper right icon to display the palette's popup menu. Select a menu item to set the zoom factor to the displayed value.

### *Info Page*

Click on the Info-Tab to switch to the info page. On this page you can change the name and position of the selected object. When two or more objects are selected it is not possible to a value. When an image is selected the size can also be changed.

## The Datatypes Palette

The Datatypes-Palette (middle right) shows the existing datatypes which can be used to build the database-tables.




The Datatypes Palette

### *Common Datatypes Page*

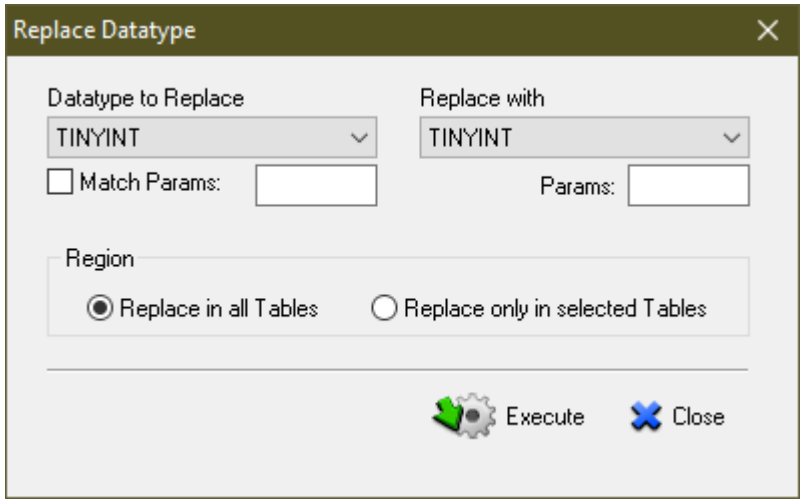
On the Common-Tab the most used Datatypes are visible and are ready to be dragged onto table columns. To add a new User Defined Datatype click the upper right icon to display the palette's popup menu. Select Create New Datatype and the Datatype-Dialog will be displayed showing the new created datatype. To remove a Datatype from the Common-Tab select Del from Common Datatypes from the palette's popup menu.

### *All Datatypes-Page*

Click on the All types-Tab to display all available datatypes. The datatypes are split into five categries. Numeric Types, Date and Time Types, String Types, Blob and Text Types and User defined Types. Doubleclick on a category to show all assigned datatypes. New created datatypes will always be assigned to the User defined Types. Doubleclick on a datatype to call the Datatype Editor. Select and right click on a datatype and select Add to Common Datatypes to add a Datatype to the Common-Tab.

### *Replace Datatypes in Tables*

If you want to replace a datatype or change a parameter in several tables then you can use the Replace Datatypes in Tables function from the palette's popup menu. It works just like the Search and Replace function in Word®.




The Replace Datatype Dialog

Select the datatype you want to replace. Then select the datatype which should be inserted. To change only the parametes of a datatype select the same datatype in both dropdowns and check the Mach Params flag and enter the parameter you want to replace and the parameter which should be inserted.

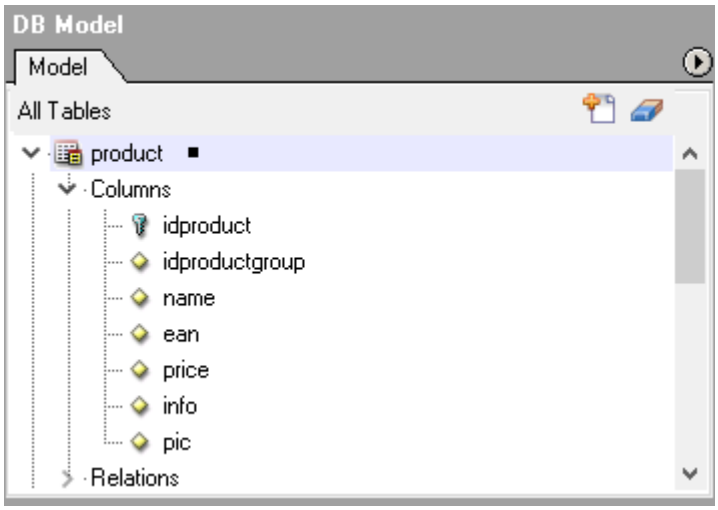
Select the Region of the function and press Execute to replace the datatype.

### *Reinitialise Datatypes*

To reset the datatypes to their initial state select Reinitialise Datatypes from the palette's popup menu. The inital settings will be loaded from the ini-files. Manually created datatypes will be replaced by the standard datatype of the model.

## The DB-Model-Palette

The DB-Model-Palette displays all existing tables in the model and their columns and relations.

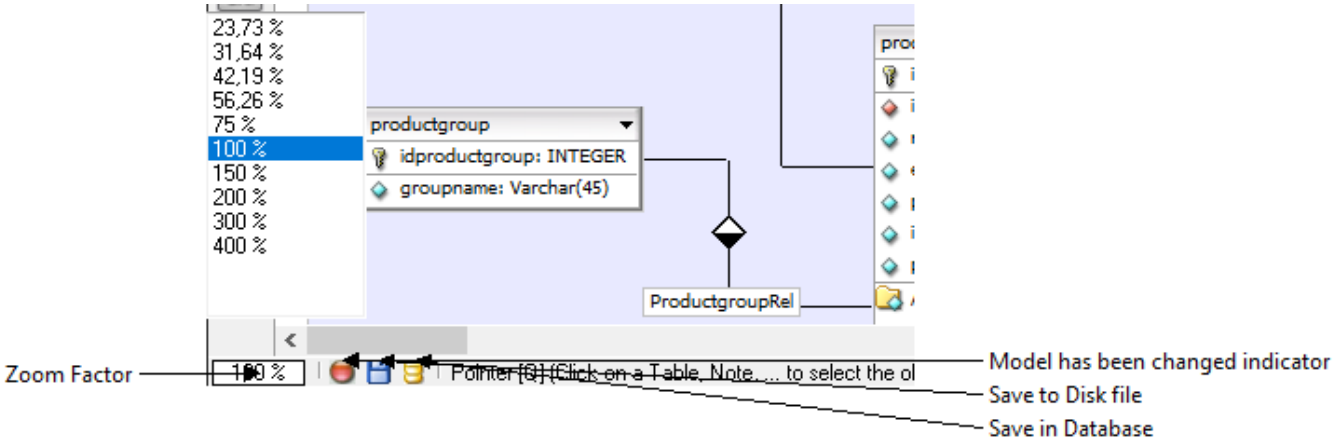



The Model Palette

Click on the plus symbols to display the columns and relations of a table. Doubleclick on a table to call the Table Editor or select Edit Object from the table's popup menu.

## The Staturbar

The Statusbar displays information about the currently selected Tool an can be used to read and set the zoom factor to a predefined value, fast.




The Statusbar

Click on the Zoom Factor Edit left on the Statusbar. A list of zoom factors pops up. Select the desired zoom factor or press ESC to close the list.

If the database model is modified the indicator left to the Zoom Factor Edit is painted red. Click on the Disc Icon to save to model to disk. Click on the Database Icon to save the model in a Database.

# Editors

Editors are used to specify and modify parameters of objects, datatypes, ...

An object's editor is usually called by doubleclicking on the object.

## Table Editor

Call the Table Editor by doubleclicking on a table or selecting [Edit Object] from the table's popup menu. You can also doubleclick the tables name in the DB-Model Palette or use the right mousebutton to call the popup menu there.

### How to use the Table Editor

To use the Table Editor the most efficient way use the following procedure.

I.) Create a new table in the model. Doubleclick on the table to call the Table Editor. The Table Name is focused so enter the table name. Press [Return].

II.) The focus will be set to the first line of the Table Columns. The first column will be named `id+tablename` automatically. If the first column has to have a different name enter the column's name. Then press [Return].

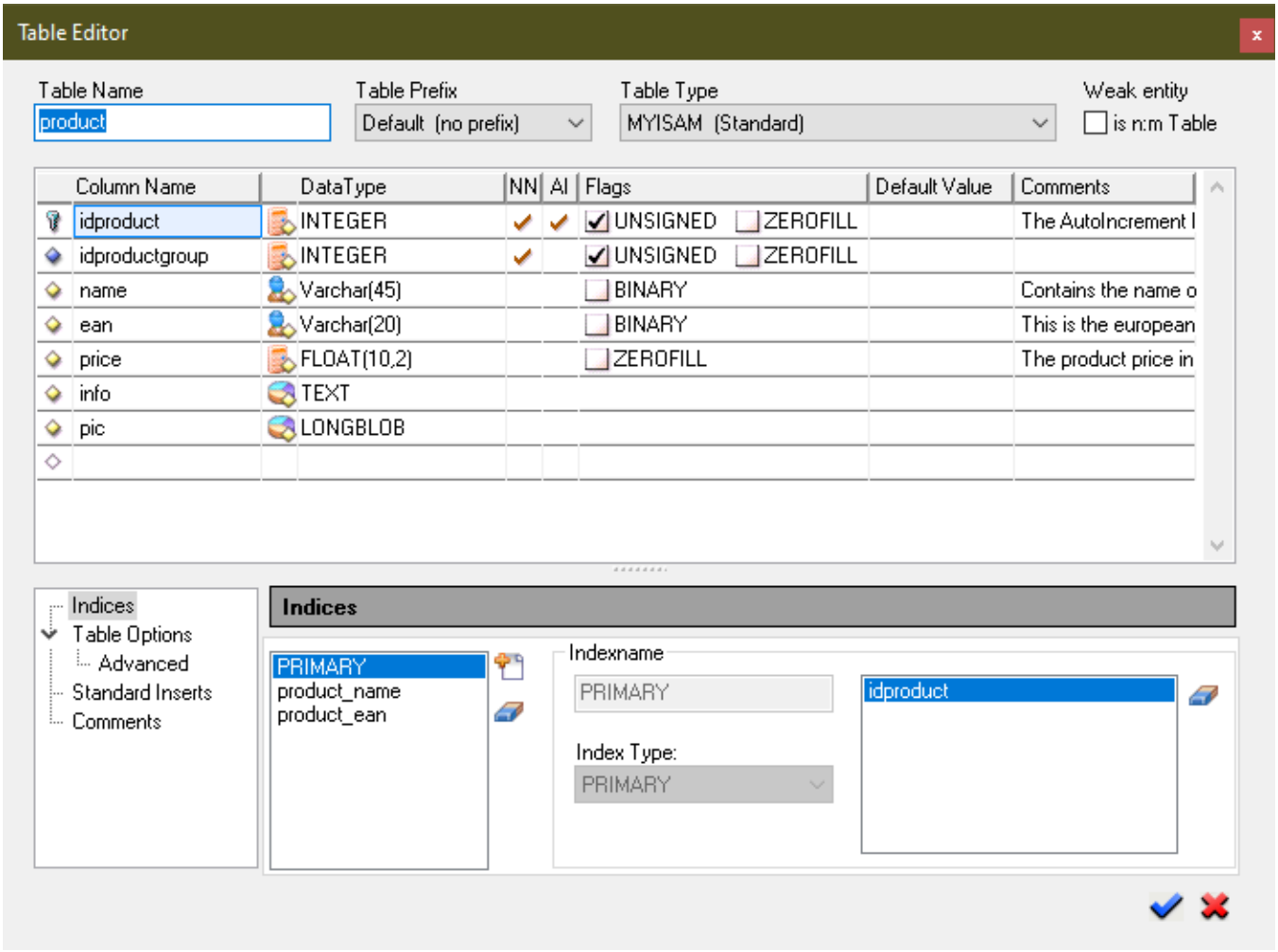
III.) Enter the next column name and press [Return]. Repeat this for all columns. Ignore the datatypes of the columns at this time, all columns' datatypes are set to the default datatype initially.

IV.) After the last column has been entered, press [Esc] to abort the column edit mode.

V.) To assign the datatypes to the columns start by dragging the first column's datatype from the Datatype palette onto the first column. Enter the datatype's parameters if the Parameter Editor pops up, e.g. VARCHAR(___)
Repeat this for all column types.

VI.) To define the primary key click on the columns' icon (left to the column name) to add or remove the column from the primary key.

VII.) Optionally define additional indices, the table options, the standard inserts and the table comments.




The Table Editor

## Table Settings

### *Table Name*

Table Name displays the current name of the table. By pressing Return or leafing the edit field the Table Name is checked against the reserved word list. If the Table Name equals a reserved word it is changed automatically.

### *Table Prefix*

In MySQL the Table Prefix can specify the database (`db_name.tbl_name`). Use this function if you want to administrate several databases in only one model. The Table Prefixes are defined in the Model Options.

### *Table Type*

MySQL supports several different Table Types. The most common types are MyISAM which is the default type and InnoDB which supports transaction-safe tables with row locking.

### *Weak entity*

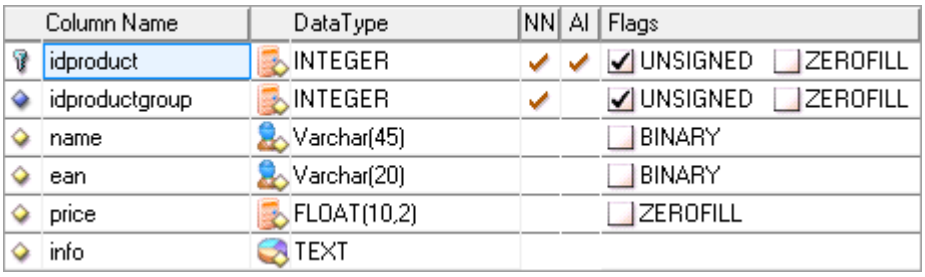
Check the [is n:m Table] checkbox if this table is a n:m Table.

## Table Columns

The cursor keys can be used to move the Cursor between the Column Name, Datatype and the Default value and the table columns. Press enter to change the focused value.

Use the mouse to dray an new datatype from the Datatype Palette onto the column. Doubleclick the datatype to change the datatype's parameters.

Click on the Table Column's row to change the NOT NULL Flag, the Column Options and the Autoincrement Flag.




The Table Columns

To enter a new column click on any column name an press the Page Down Key. The Cursor is placed below the last column. Add the new column by entering the column name.

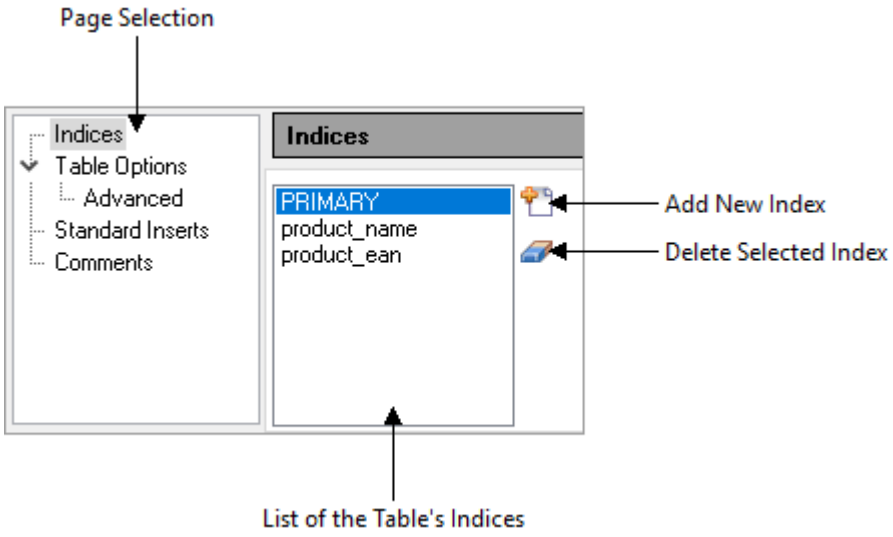
If you want to enter several new columns press return. The Curor is moved to the next row and the next column name can be entered.

Drag the appropriate datatype from the Datatype Palette onto the new column. Enter the datatype's parameters if needed.

Set the Column's NOT NULL Flag, the Column Options and the Autoincrement Flag if nessesary.

## Table Indices

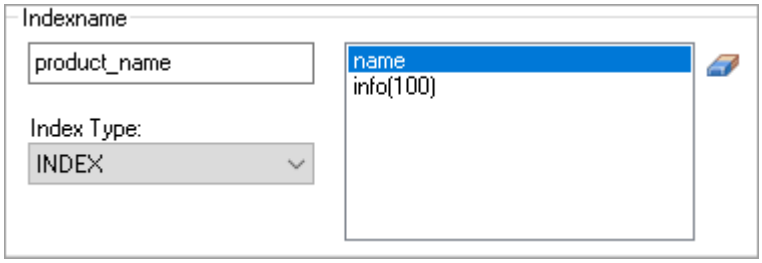
An unlimited number of indices can be defined for each table. The PRIMARY index is defined automatically and cannot be deleted.




The Indices List

To add a new index click the plus button and enter the index name. Select the Index Type. You cannot select the primary type manually because there can only be one primary index.

To delete an index select it in the index list and click the rubber button.




An Index

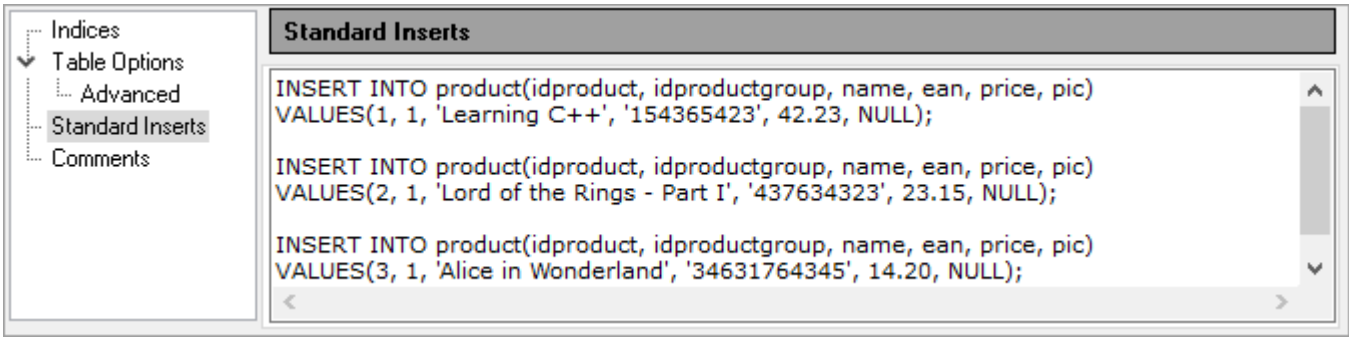
To add columns to the index drag the appropriate Table Columns down to the Index Columns listbox. You can reorder the index's columns by drag'n'drop in the Index Columns listbox.

To delete a column from the index select the column in the Index Columns listbox and click the rubber button to the right.

## Standard Inserts

Every table can have it's Standard Inserts. They provide initial data for the tables.

When the SQL Creates are exported or the database is created by the syncronise function the Standard Inserts can be exported or executed as well.



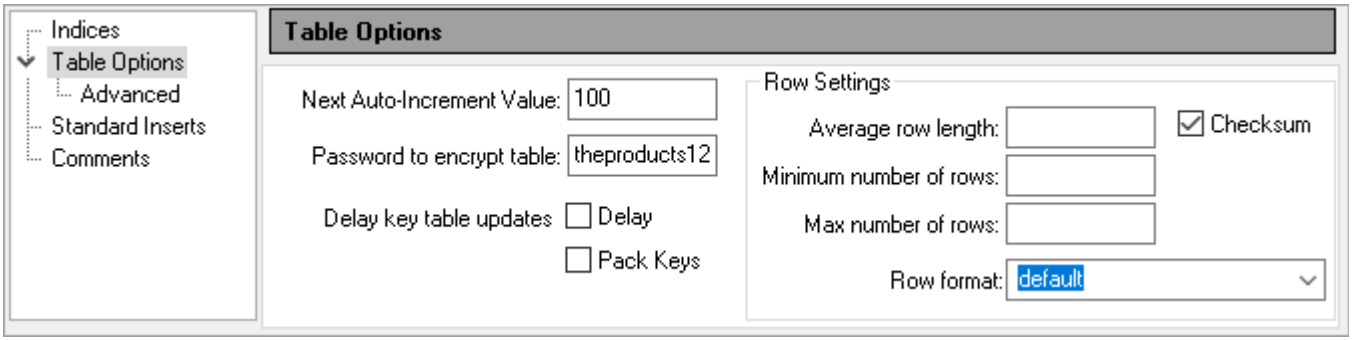

The Standard Inserts

To add a SQL INSERT command click the right mouse button and select Paste SQL Insert from the Standard Inserts memo's popup menu.

To clear the Standard Inserts click on the rubber button right to the Standard Inserts memo.

## Table Options

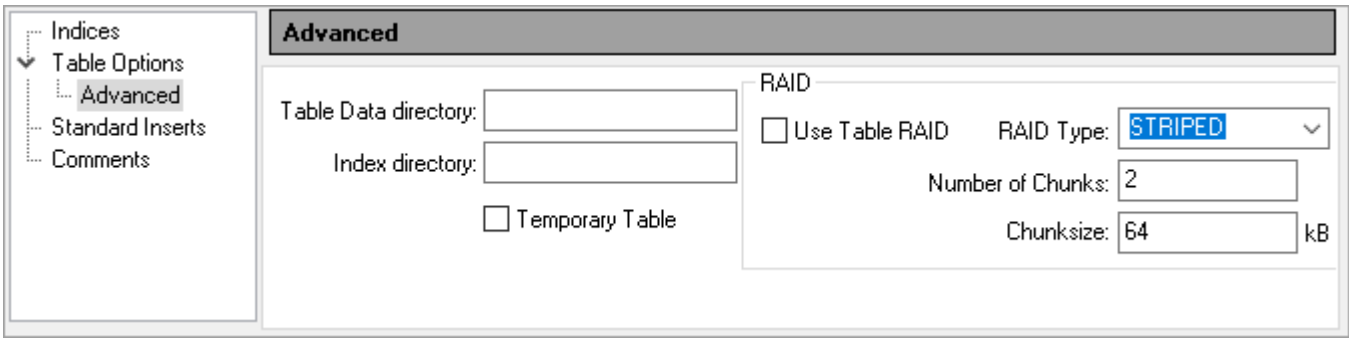
The Table Options provide access to the MySQL specific table options. Please refere to the MySQL documentation to understand the functionality of the several Table Options.




The Table Options

## Advanced Table Options

Like the Table Options the Advanced Table Options provide access to the MySQL specific table options. Please refere to the MySQL documentation to understand the functionality of the several Table Options.




The Advanced Table Options

## Comments

To store additional information about the table select the Comments and enter the information.

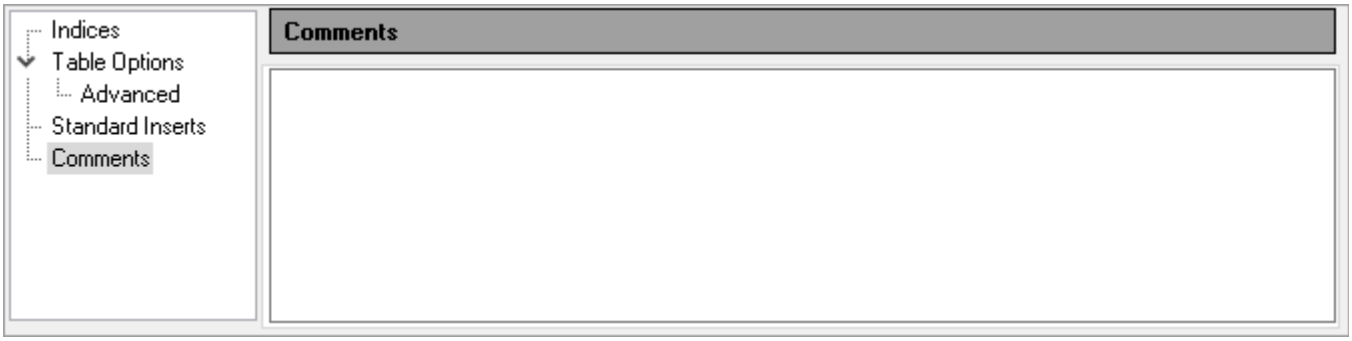
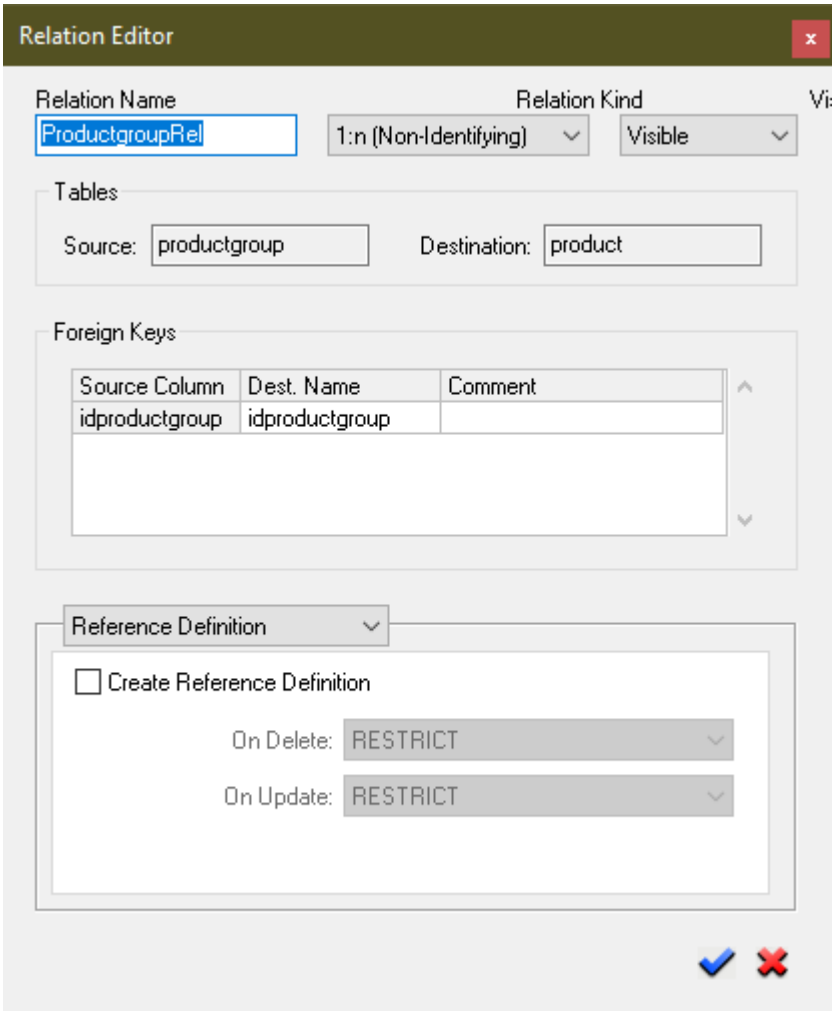



Table Comments

# Relation Editor

The Relation Editor allows you to change the Kind of the relation, the Foreign Keys fields mapping and the Relation Reference Definition.




The Relation Editor

### *Relation Name*

Change this value to rename the relation.

### *Relation Kind*

A relation can be one of the following kinds.

| Kind | Caption | Explanation | Example |
|---|---|---|---|
| 1:1 | one - to - one relation | One row in the source table matches one row in the destination table. | A relation between a [customer] table and an [address] table. One customer has one address. |
| 1:n | one - to - many relation | One row in the source table matches many rows in the destination table. | A relation between an [order] table and a [orderproduct] table. One order can have many products and a product in the [orderproduct] table is only part of one order. |
| 1:n (Non Identifying) | one - to - many relation, FK not in PK | One row in the source table matches many rows in the destination table but the Foreign Key in the destination table is not in the Primary Key Index. | A relation between a [payment] table and an [order] table. Each row in the [order] table as a [payment] assigned to. Only [idorder] is the Primary Key field in the [order] table. |
| n:m | many - to - many relation | A n:m relation is always broken down into two 1:n relations. | |
| 1:1 (Generalisation) | one - to - one relation | This relation is treated like a normal 1:1 relation within DBDesigner Fork. Some plugins make use of this type of relation. | |

### *Invisible*

Use this option to hide the relation. To make the relation visible again select the source table or the destination table in the Model Palette, expand the relations and doubleclick the invisible relation. In the Relation Editor disable this option.

### *Foreign Keys*

Use the Foreign Key table to change the fieldnames of Foreign Keys in the destination table and to add additional comments to the fields.

Doubleclick a value to change it.

### *Reference Definitions*

Select the Create Reference Definition checkbox to activate the Reference Definitions. Use the comboboxes to select the appropriate actions for delete and update events.

Please note that native MySQL tables do not support Foreign Key Reference Definitions. Use InnoDB if Foreign Key Reference Definitions are needed.

### *Optional Relation*

To make the relation optional on one side check the appropriate option.

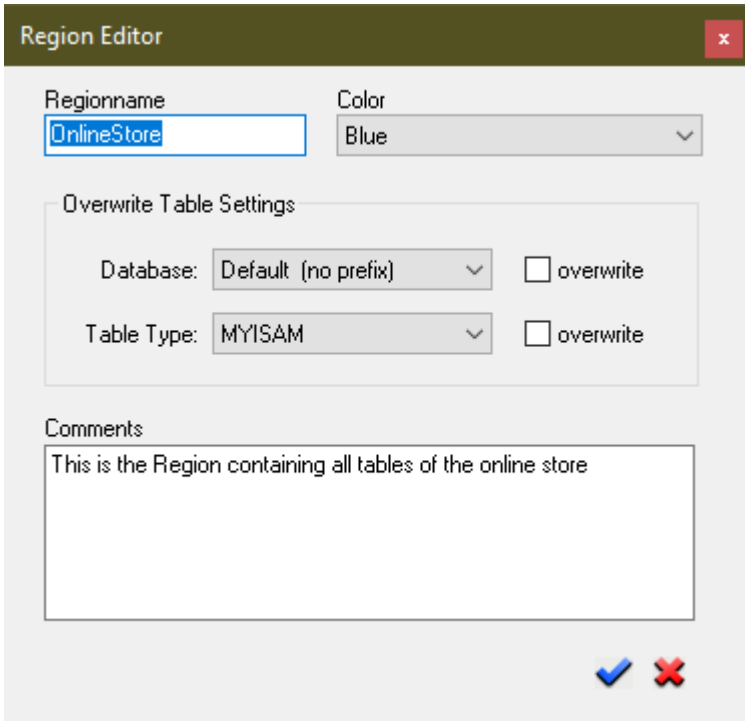
### *Comments*

Use the Comments Memo to enter additional information about the relation.

## Region Editor

Call the Region Editor from the region's popup menu. The Region Editor allows you to define default table settings for all tables positioned on the region.

To activate set default settings selected the wanted options and enable the checkbox next to that option.




The Region Editor

### *Regionname*

Change this value to rename the region.

### *Color*

Each region is painted in an individual color. Choose from the list to specify the region's color.

### *Overwrite Table Settings*

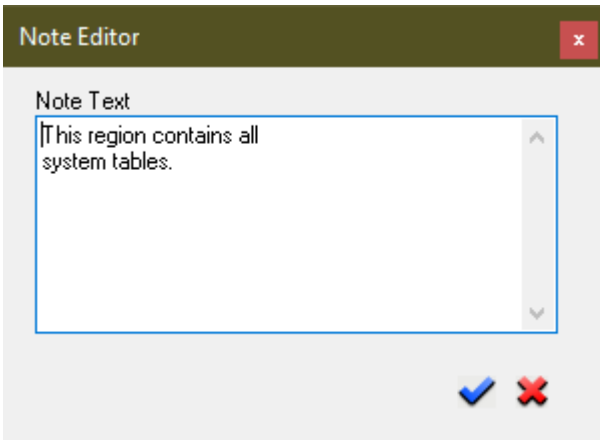
Select the appropriate settings and check the overwrite checkbox to activate them.

### *Comments*

The comments memo can be used to store informations about the region.

## Note Editor

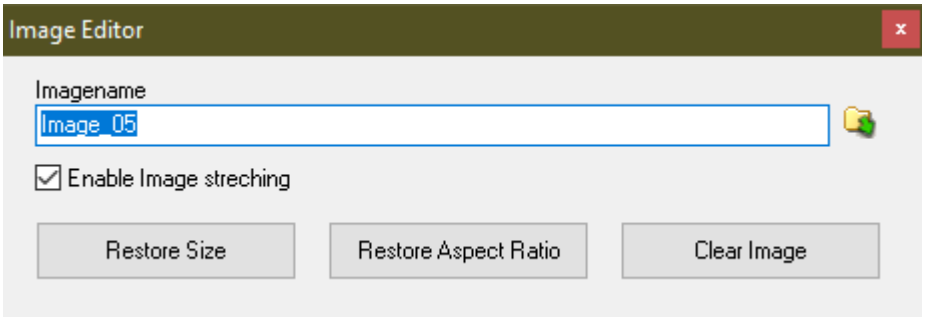
Doubleclick on a new created note to enter call the Note Editor and enter the note's text.




The Note Editor

## Image Editor

Doubleclick on a image to call the Image Editor.




The Image Editor

### *Load New Image*

To replace this image with an updated version of the image or a different image click the Image Folder button. A file browser is shown and you can select the image.

Currently only the PNG and BMP image formats are supported.

### *Strech Image*

When the Strech Image option is selected you are able to resize the image.

### *Restore Size*

To restore the original size of the image after it has be rescaled press the Restore Size button.

### *Restore Aspect Ratio*

When the image has been resized and the aspect ratio was changed press the Restore Aspect Ration button to change the height of the image according to the aspect ration.

### *Clear Image*

Press the Clear Image button to remove the bitmap.

# Datatype Editor

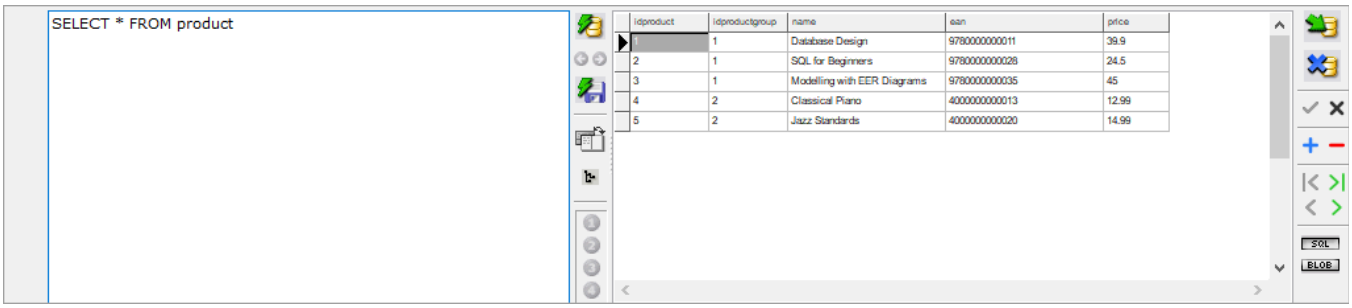
Doubleclick on a datatype in the Datatype Palette to call the Datatype Editor.

# Query Editor

To call the Query Editor from Design Mode right-click on the table and select Edit Table Data from the popup menu.

When DBDesigner Fork is Query Mode a docked Query Editor is shown at the bottom of the main window. To open another editor double-click on the appropriate table.

If there is no active database connection the Database Connection Dialog will be shown. Create a new or select an existing database connection and click connect.




The Query Editor

### *Main areas*

The Query Editor can be divided into two main areas, the SQL Command Editor and the Data Grid.

The SQL Command Editor is used to specify a SQL Command. The Data Grid will display the result returned from the Database.

The SQL Command Editor is extended by the SQL Command Storage Tree. The SQL Command Storage Tree is used to store SQL commands permanently with the model and to keep a command history.

The Data Grid is extended by the BLOB Editor. BLOB Editor is used to modify, load and save data from or to BLOB fields.

### *Viewing and hiding the areas.*

When the Query Editor is opened from Design Mode or by double-clicking a table in Query Mode, only the Data Grid is visible. To display the SQL Command Editor click the SQL button at the right.

To display the SQL Command Storage Tree click on the tree icon right to the SQL Command Editor.

To edit the content of a BLOB Field click on the BLOB button at the right.

# Database Creation and Maintenance

## Traditional SQL Export

Like any other database modelling tool DBDesigner Fork can export the model as a SQL script file which can be executed by any database maintenance tool, like the command line tools mysql, sqlite3 or isql of Firebird.

All SQL CREATE TABLE statements and the Standard Inserts can be written to the SQL script file depending on user settings. The script can be written for MySQL, SQLite, Firebird, Oracle, PostgreSQL and SQL Server.

It is also possible to output all SQL DROP TABLE statements.

## What is Database Synchronisation?

With DBDesigner Fork you can simplify the task of creating and maintaining your database. DBDesigner Fork can connect to a MySQL, SQLite or Firebird database and create and synchronize it with the designed model.

Synchronisation means that DBDesigner Fork scans all tables in the existing database and checks for differences. If a table is present in the model but not in the database, the according SQL CREATE TABLE statement is executed.
If the table is not present in the model but in the database it can be deleted depending on user settings.
If the table exists in the model and the database as well, all fields, indices and foreign keys are compared and if there is a difference, the appropriate SQL ALTER TABLE statements are executed.

The databases differ in what ALTER TABLE can do:

- **SQLite** can rename, add and drop columns only. Every other change rebuilds the table: a new table is created, the rows are copied, the old table is dropped and the new one renamed; indices and triggers are created again. Every table is rebuilt in a transaction of its own. A rebuild is refused while `PRAGMA foreign_keys` is switched on for the connection.
- **Firebird** cannot rename a table; the table is created with the new name, the rows are copied and the old table is dropped. An existing column cannot be turned into an identity column. Every DDL statement is committed on its own; a failing statement is reported in the Progress List and the synchronisation goes on.
- The model keeps its datatypes. For Firebird they are mapped to the types of the database (e.g. DATETIME to TIMESTAMP, TEXT to BLOB SUB_TYPE TEXT, AUTO_INCREMENT to an identity column).

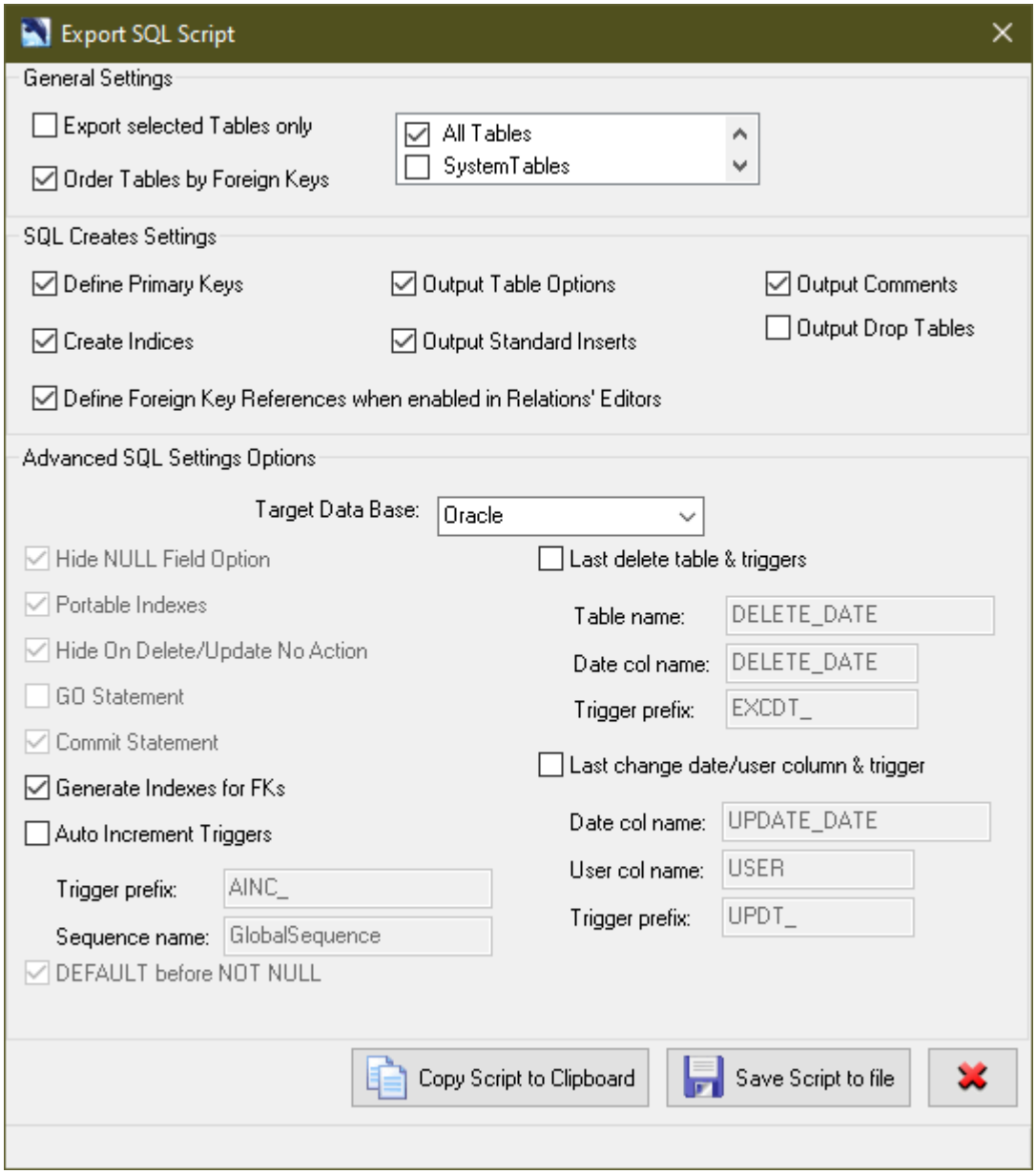
## What is Reverse Engineering?

To reverse engineer a database means to connect to a database server, take an existing database and automatically build a database model based on the meta information in the database.

Within DBDesigner Fork all table information is extracted from the meta information: columns with their datatypes and defaults, primary keys, indices, foreign keys and comments. The tables are placed on the model in alphabetical order following a grid scheme defined by the user.
The process of reverse engineering is possible with MySQL, SQLite and Firebird databases. For SQLite and Firebird it also keeps generated or computed columns, indices on an expression, descending and partial indices, CHECK constraints and the table options of the database (e.g. STRICT and WITHOUT ROWID for SQLite, global temporary tables for Firebird), so that a following synchronisation does not change the database.

# Export SQL Creates

To export a SQL create script based on you current database model select File-Export-SQL Create Script ...
The Export SQL script dialog appears.




Export SQL Creates dialog

To export the SQL Creates to a file, press the [Save script to file] button. You will be prompted for the file's name and destination. Press [Save] to write the script to disk.

When a sql shell is open, it can be useful to copy the script to the clipboard and execute it directly from the sql shell. To copy the SQL script to the clipboard press the [Copy Script to Clipboard] button.

## SQL Create Options

The output can be customised using the following General Settings, the SQL Creates Settings and the Advanced SQL Settings. The settings are remembered for the next export.

### *Export selected tables only*

Check this flag to export the selected tables only. All other tables will not be created by the script file. The list below the flag can be used to export the tables of single regions only.

### *Order Tables by Foreign Keys*

Use this flag to change the creation order. By default the tables are created in alphabetical order. When you are using foreign keys it is necessary to change the order the tables are created.

Tables with no relations pointing to them have to be created first. Every other table can only be created when all source tables already exist.

If there is a collection of cyclic relations, the tables cannot be created. An error message is displayed. Note that you still can export the tables in alphabetical order.

### *Define Primary Keys*

Check this flag if you want to enable the creation of primary keys.

### *Create Indices*

Check this flag if you want to enable the creation of indices. This does not include the primary keys. Check the Define Primary Keys flag to create primary keys.

### *Define Foreign Key References when enabled in Relations' Editors*

Use this option to enable the foreign key reference in the SQL CREATE TABLE statements. A reference is written for every relation whose Reference Definition is activated in the Relation Editor. Note that you have to enable the Order Tables by Foreign Keys option to make the SQL script work.

### *Output Table Options*

Use this option to enable the table options in the SQL CREATE TABLE statements. This is only necessary if you have specified Table Options for a table in the model.

### *Output Standard Inserts*

Use this option to export the Standard Inserts within the SQL create script. This is only necessary if you have specified Standard Inserts for a table in the model.

### *Output Comments*

Writes the comments of the tables and columns into the script, in the form the target database uses (e.g. COMMENT for MySQL, COMMENT ON for Firebird).

### *Output Drop Tables*

Writes a DROP TABLE statement in front of every CREATE TABLE statement, so the script can be run against a database which already contains the tables.

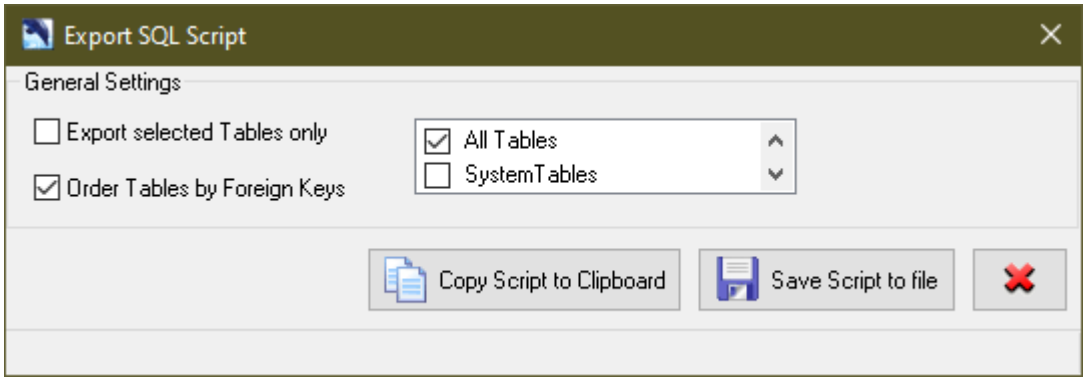
### *Advanced SQL Settings*

Select the **Target Data Base** first: FireBird, My SQL, Oracle, PostgreSQL, SQL Server or SQLite. The datatypes, the quoting of names and the form of the statements follow the target. Some of the other options are fixed by the target and are shown disabled:

- **Hide NULL Field Option** and **Portable Indexes** are used for every target except MySQL.
- **Hide On Delete/Update No Action** leaves out these rules for Oracle, PostgreSQL and SQL Server.
- **Generate Indexes for FKs** creates an index for every foreign key; it is preselected for Oracle, PostgreSQL and SQL Server.
- **GO Statement** (SQL Server) and **Commit Statement** (Oracle, PostgreSQL, Firebird) separate the statements.
- **DEFAULT before NOT NULL** is the order of the column options Oracle, PostgreSQL and Firebird expect.
- **Auto Increment Triggers** (Oracle, Firebird) write a sequence or generator and a trigger for every auto increment column. Without this option an auto increment column becomes an identity column for Firebird.
- **Last change date/user column & trigger** and **Last delete table & triggers** (Oracle, Firebird, SQL Server) add columns and triggers which record the time and the user of the last change and of deleted rows. The names of the columns, tables and triggers can be entered next to the options.

## Export SQL Drops

To export a SQL drop script based on you current database model select File-Export-SQL Drop Script ... The Export SQL script dialog appears.




Export SQL Drops dialog

To export the SQL Drops to a file, press the [Save script to file] button. You will be prompted for the file's name and destination. Press [Save] to write the script to disk.

When a sql shell is open, it can be useful to copy the script to the clipboard and execute it directly from the sql shell. To copy the SQL script to the clipboard press the [Copy Script to Clipboard] button.

## SQL Drop Options

The output can be customised using the following General Settings.

### *Export selected tables only*

Check this flag to export the selected tables only. All other tables will not be dropped by the script file.

### *Order Tables by Foreign Keys*

Use this flag to change the order. By default the tables are dropped in alphabetical order. When you are using foreign keys it is necessary to change the order the tables are dropped.

The tables will be dropped the reversed order they have been created.
If there is a collection of cyclic relations, the tables cannot be dropped in this order. An error message is displayed. Note that you still can export the tables in alphabetical order.

# Database Connections

Several functions in DBDesigner Fork use Database Connections: the Database Synchronisation, the Reverse Engineering, the Query Mode, the storage of models in a database and the plugins. They are used to establish a connection to a database by selecting the appropriate server and database.

Connections can be made to **MySQL 8**, **SQLite 3** and **Firebird 3** or newer databases. The drivers Oracle, MSSQL and ODBC are still listed, but there are no connectors for them in this version.

### *Create a new Database Connection*

A new Database Connection is created in the Database Connection Dialog.

To create a new Database Connection, click the New Database Connection button. The Connection Parameter Dialog is shown. Specify all needed information and click OK to add the connection to the connection list.

### *Connect to a database*

Like creating a new Database Connection, the Database Connection Dialog is used to connect to a database.

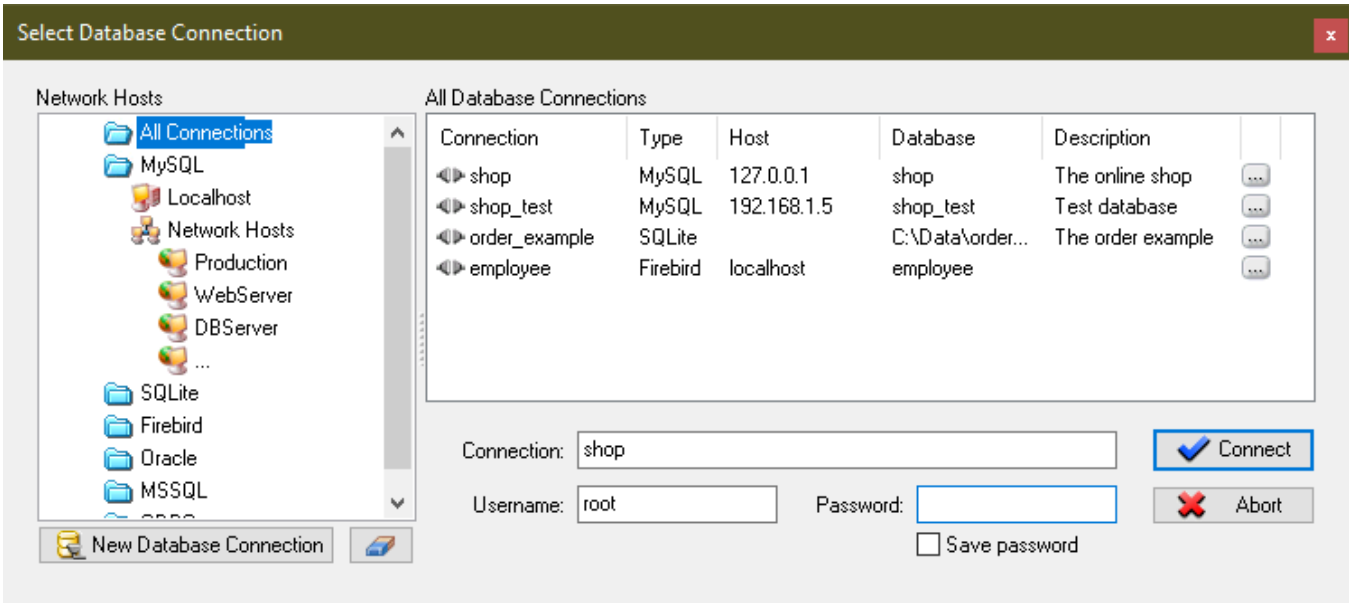
Select the appropriate connection from the connection list. Enter a username and the password and press the Connect button to establish the connection.

### *Please notice*

To make changes to meta information in a database it is necessary to connect as an user with the required rights. Within MySQL the database administrator is called root, within Firebird SYSDBA. The administrator has got all rights and is allowed to create a new database or make changes to any database. Every other user is allowed to access and modify only the tables he has got the appropriate rights for. SQLite has no users; the rights on the database file count.

To avoid problems while synchronizing or querying a database make sure the user which is used to build the connection to the database has the required rights.

## The Database Connection Dialog




Database Connection Dialog

The Database Connection Dialog is seperated into three areas, the Network Hosts Tree, the Connection List and the User/Password Section. The dialog can be resized.

## Network Hosts Tree

The Network Hosts Tree displays all entered hosts and their databases. It is used to filter the displayed connections and to create new connections to a host's databases.

When the first node named [All Connections] is selected, all entered connections are displayed in the Connection List.

To display all connections to the local MySQL host select [MySQL]-[Local Host].

To display all connections to MySQL host located on the network select [MySQL]-[Network Hosts].

To display all connections to a specific MySQL network host select the hosts name beneath the [MySQL]-[Network Hosts] node.

Connections to SQLite and Firebird databases are listed beneath the [SQLite] and [Firebird] nodes. Click on the node below them to create a new connection of this type.

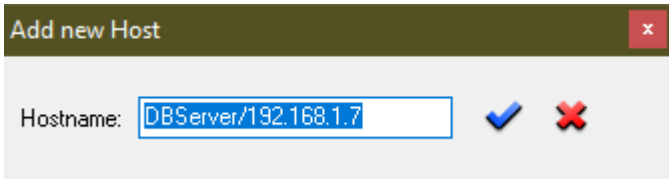
### *Display a Host's databases*

To display the databases of a entered MySQL Host click on the [+] icon left to the Host's name. DBDesigner Fork prompts for the user's name and password which will be used to log in. Note that the specified user has to have the appropriated rights to execute a SQL SHOW DATABASES command.

### *Enter a new Host*

After the installation of DBDesigner Fork it is possible only to connect to the local MySQL host. To add a new network host obey the following.

To enter a new Host, click onto the [...] node listed beneath the [Network Hosts] node beneath of the wanted database type. The Add new Host Dialog will apear.



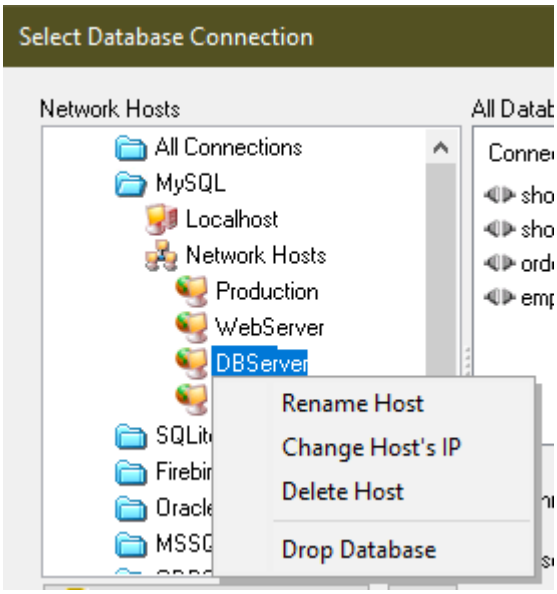

Add new Host Dialog

Enter the node's name followed by a slash and it's IP address. In place of the IP address the host's network name can be specified also, e.g. webserver or www.theserver.com.

Press return to append the new Host to the Network Hosts Tree.

### Change Host parameters

To change a Hosts name or IP address right click on a Host. The Host's popup menu will be displayed.




Host popup menu

Select the desired function from the menu.

### Remove a Host

To delete a Host select [Delete Host] from the popup menu.

### Create a new database

It is possible to create a new MySQL database from within the Database Connection Dialog. To create a new database display the Host's databases like descibed above. Click the last node below the Host's node which is labeled [...]. The New Database Dialog will apear. Enter the database name and press return. The database will be created.

SQLite and Firebird databases are files. A connection to a SQLite database file which does not exist creates the file. A Firebird connection without a host name uses the embedded engine and creates a missing database file, too.

### Drop a database

It is possible to drop a MySQL database from within the Database Connection Dialog. To drop a database show the Host's databases like descibed above. Click the database's node with the right mouse button to display the popup menu. Select [Drop Database].

Be aware that once a database is dropped it cannot be restored. A database backup has to be restored instead.

## Connection List

The Connection List displays the connections selected in the Network Hosts Tree. Click on the wanted connection to set the connection in the User Section.

### *Creating a new connection*

To create a new connection select the Host to connect to in the Network Hosts Tree. Display the Host's databases. Now drag the database to connect to onto the Connection List. A new connection is created.

Instead of dragging the database you can click the [New Database Connection] button, or right click on the list and select [Add Connection].

### *The list's columns*

The list has got five columns: Connection, Type, Host, Database and Description. Double click on a connection or select [Edit Connection] from the popup menu to change its parameters in the Connection Parameter Dialog. Select [Delete Connection] to remove it from the list.

## The User Section

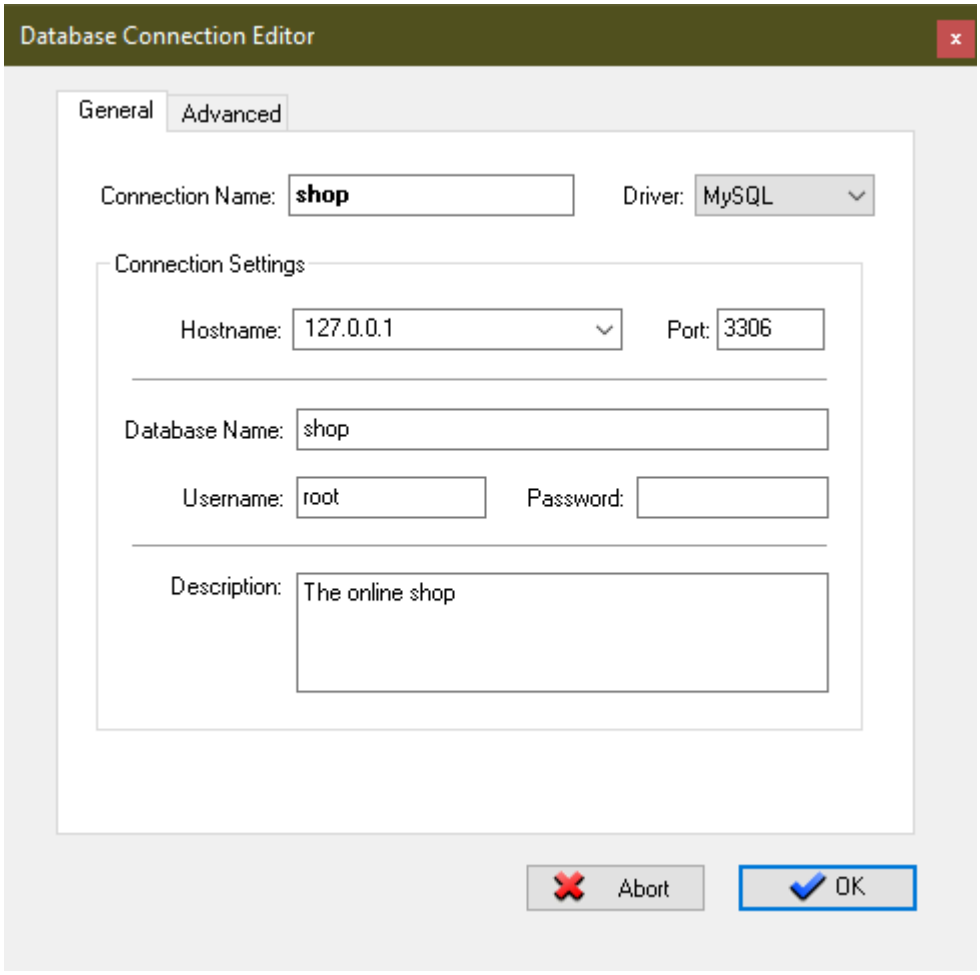
When a database connection is selected in the Connection List the connection's name is displayed in the User Section and DBDesigner Fork prompts for the user's password.

On Windows the password can be stored with the connection: check [Save password] before you connect. The password is encrypted for the Windows user account (Windows Data Protection API) and can only be read by this user on this computer. Without the check mark the password is never stored. On Linux the option is not available.

Press enter or click the [Connect] button to establish the database connection. If the connection was successfully build the connection dialog is closed. If an error occured, e.g. the password was wrong, the error message is displayed.

## Connection Parameter Dialog

The Connection Parameter Dialog (Database Connection Editor) is used to enter or change the parameters of a connection.




Connection Parameter Dialog

### *Connection Name*

Each Database Connection is identified by a unique name.

### *Driver*

Select a database driver from the dropdown list: MySQL, SQLite or Firebird. When a new database driver is selected all values are set to their initial state.

### *Hostname and Port*

Enter the database server's IP address or network name and its port. They are needed for MySQL (port 3306) and Firebird (port 3050). Leave the host name of a Firebird connection empty to use the embedded engine of the client library; no Firebird server has to be installed then. SQLite has no server, the fields are disabled.

### *Database Name / Database File*

When using the MySQL driver enter the name of the database, the name which was used in the CREATE DATABASE SQL statement.

When using the SQLite driver enter the path of the database file.

When using the Firebird driver enter the path of the database file on the server (or on this computer for the embedded engine), or an alias defined on the server.

### *Username*

Specify the username used to connect to the database. SQLite does not need a username.

### *Password*

Specify the password used to connect to the database.

### *Description*

Enter a description with a short information about the database connection.

### *Advanced Tab*

Beware. Only modify the advanced connection parameters if you know what you are doing. The [Reset to Defaults] can be used to reset the connection parameters to their defaults. To add or delete a parameter use the according buttons.

The parameter `VendorLib` names the client library of the database. For Firebird it can be the full path of `fbclient.dll`; otherwise the library is looked for next to the program and then in an installed Firebird server.

The default parameters can be modified by editing the file DBConn_DefaultSettings.ini located in the DBDesigner Fork data directory.

## Database Synchronisation

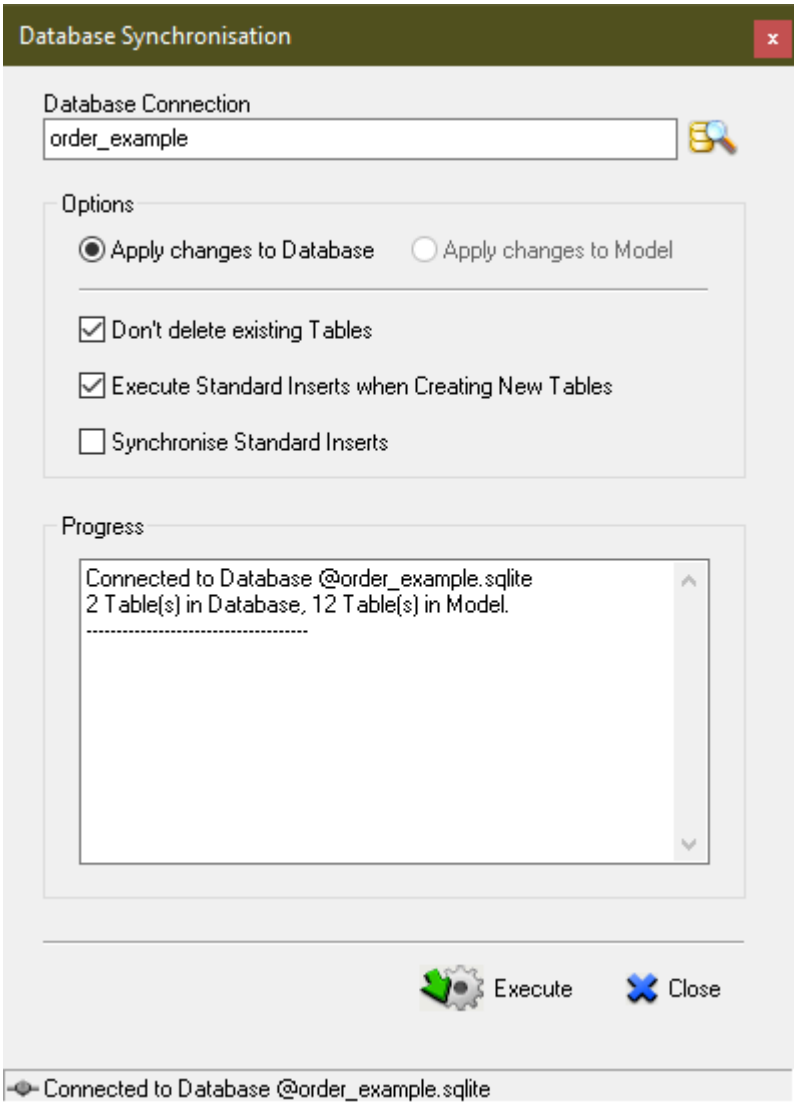
To synchronise the model with a database select [Database]->[Database Synchronisation] from the main menu. The synchronisation can also be called by pressing the [Sync] button in the Toolbar. Note that an empty model cannot be synchronised.

The Database Connection Dialog will be shown. Select the connection to the database which should be synchronised. Enter the user's password and press [Connect] to establish the connection.

### *Synchronise with a new database*

The synchronise function can be run against an empty database. All tables will be created and the Standard Inserts will be inserted into the tables.

To synchronise the model with a new MySQL database call the Database Synchronisation. The Database Connection Dialog is displayed. To create a new database display the Host's databases and click the node labeled [...] like explained above. Name the database an create a new connection by dragging the database's node onto the Connection List. Connect to the new database. For SQLite and Firebird create a connection to a database file that does not exist yet.



Database Synchronisation

## Synchronisation options

When the connection to the database is established successfully the Database Synchronisation Dialog is displayed.

### *Database Connection*

The name of the database connection is displayed at the top. To connect to a different database click the button right to the connection's name. The Database Connection Dialog is displayed again and a different connection can be selected.

### *Apply changes to Database*

This option is seleted by default. The database will be modified to reflect the changes made to the model. The model will be unchanged when the synchronising function is executed.

### *Apply changes to Model*

This option is disabled. To take changes of the database into the model use the Reverse Engineering function instead.

### *Don't delete existing Tables*

Check this option to keep tables in the database which are not present in the model. If this option is not checked these tables will be dropped.

### *Execute Standard Inserts when Creating New Tables*

If this option is activated the Standard Inserts will be inserted into the new created table. This option is selected by default.

### *Synchronise Standard Inserts*

Activate this option when changes to the Standard Inserts of a table have been made. The rows will be compared using the primary key fields. If a row in the database matches with a row in the Standard Inserts all the values entered in the Standard Inserts will be compared. Values not present in the Standard Inserts will be ignored. Appropriate changes are made to the database table rows.

Database table rows with a primary key value which is not matched by and primary key value in the Standard Inserts are kept unchanged. Because of that manually entered rows will stay in the table even when this option is used. If a Standard Insert is deleted in the model the according row in the database table has to be deleted manually.

## Executing the synchronisation

The synchronisation cannot be undone. Before executing check all selected options. Check the Progress List which will display the connection info and the number of tables in the database and the model. The connection info will be printed like this [user@database], e.g. [root@webshop].

To execute the synchronisation press the [Execute] button at the bottom of the dialog.

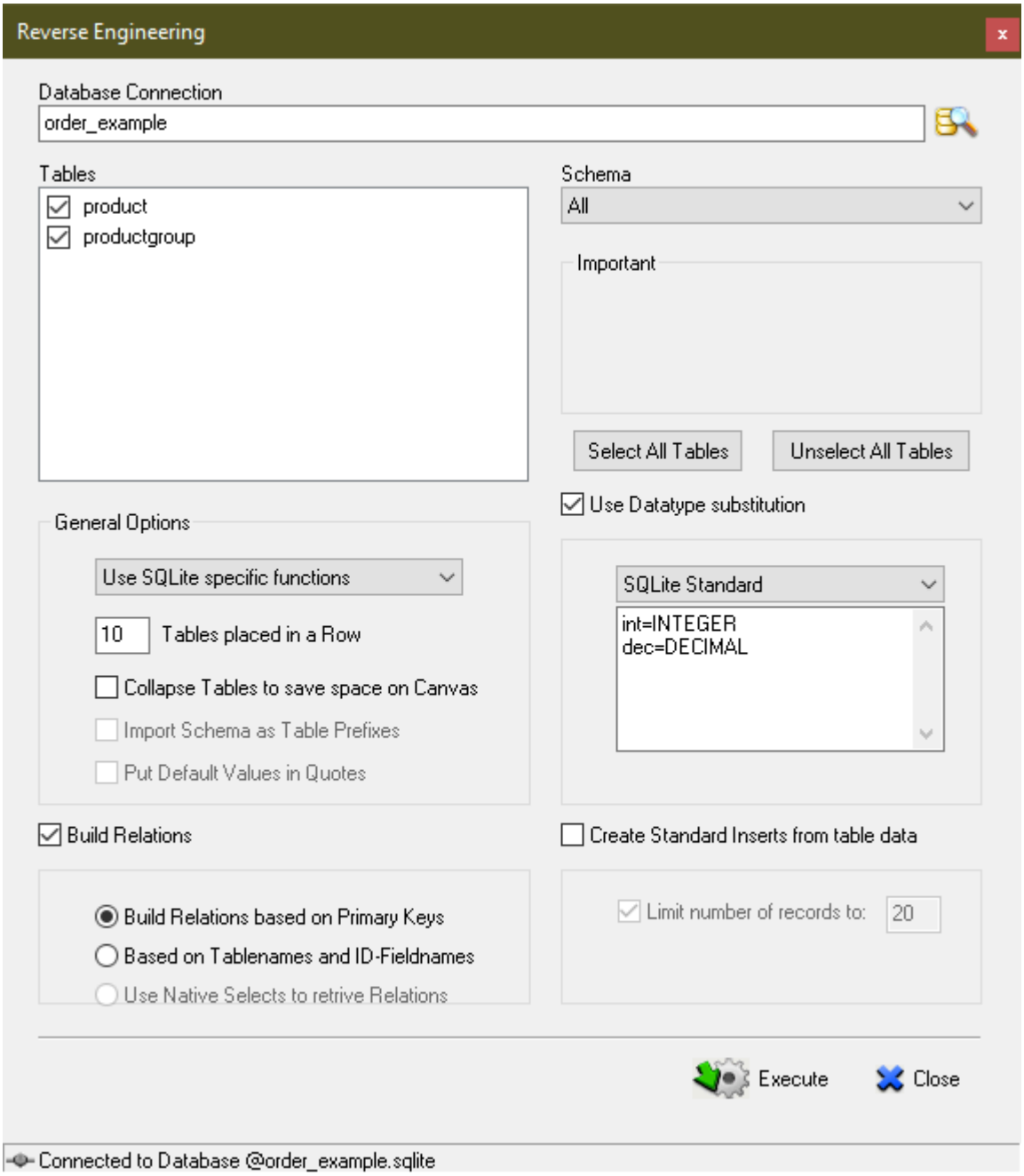
The execution can be monitored through the Progress List. All checks and changes are listed. If the synchronisation is aborted by an error, the error message is shown and written to the Progress List.

# Reverse Engineering

To call the Reverse Engineering function select [Database]->[Reverse Engineering] from the main menu. The function can also be called by clicking the [Rev.] button in the Toolbar.

Normally the Reverse Engineering function is called with an empty model. It is also possible to add the tables to an existing model. If the database tables should be created within a new model select [File]->[New] from the main menu before calling the function.

The Database Connection Dialog will be show. Select or create a new connection to the database which should be reverse engineered and establish the connection.




Reverse Engineering Dialog

When the connection to the database is established successfully the Reverse Engineering Dialog is displayed.

### *Database Connection*

The name of the database connection is displayed at the top. To connect to a different database click the button right to the connection's name. The Database Connection Dialog is displayed again and a different connection can be selected.

### *Tables*

Select all database tables which should be created in the model. All tables are selected by default. To select all tables when some of them have been deselected press the [Select All Tables] button. To deselect all tables press the [Unselect All Tables] button.

## General Options

### *Database specific functions*

The dropdown list selects how the meta information is read. It is set according to the driver of the connection: Use MySQL specific functions, Use SQLite specific functions or Use Firebird specific functions. They read the

most accurate copy of the table's structure. Do not change this setting.

### *Tables placed in a Row*

When the database is reverse engineered all the selected database tables are placed in a grid on the models canvas. This option specifies the number of tables in a row.

The options Import Schema as Table Prefixes and Put Default Values in Quotes are only used for Oracle databases.

## Build Relations

Check Build Relations to let DBDesigner Fork create the relations between the tables. First a relation is created for every foreign key of the database, with its ON UPDATE and ON DELETE rules; a foreign key which refers to its own table becomes a relation of the table to itself. The following options add relations for columns which have no foreign key in the database. Please note that this may lead to incorrect results. So always check the relations after the database has been reverse engineered.

### *Build Relations based on Primary Keys*

The relation will be built upon the table primary key fields. When the PK fields of a table are present in another table a 1:n relation is created.

### *Based on Tablenames and ID-Fieldnames*

The relation will be built upon the table's and primary key field's names. The following naming principle is assumed. The primary key of a table is named ID + Tablename. If the primary key name is present in a different table a one-to-many between the two tables is created.

## Use Datatype substitution

Check this option if you want to change some datatypes' names. This option can be used to map different datatypes between different databases.

By default the Datatype Substitution [MySQL Standard] is selected. This substitution will replace MySQL's abbreviations [int] and [dec] with the [INTEGER] and [DECIMAL] datatypes.

To change a datatype's name enable this option, change the Datatype Substitution to [User defined] and add a line to the Datatype Substitution List. Use the following format. [OldDataypeName=NewDatatypeName], e.g. [int=INTEGER] will replace all [int] datatypes with [INTEGER] datatypes.

## Executing the function

To execute the function press the [Execute] button at the bottom of the dialog.

## Create Standard Inserts from table data

This option can be used to automatically create Standard Inserts for the created tables. The Standard Inserts will be built from the table's data stored in the database.

### *Limit number of records to...*

The number of created Standard Inserts can be limited to a maximum number by using this option. This can be useful if there are many rows stored in the table but only a few records are needed for testing purposes.

# Plugins

Plugins are separate programs which work on the current model. They are started from the Plugins Menu. DBDesigner Fork saves the model to a temporary file and starts the plugin with this file. While the plugin runs, the main window is disabled. When the plugin has changed the model, the changed model is loaded again and you are asked whether it should be saved.

The plugins use the stored database connections, the language and the font of DBDesigner Fork; a connection opened in DBDesigner Fork is offered to the plugin. The following plugins are installed with the program.

## Data Browser

The Data Browser shows the rows of the tables in a connected database. Click [Connect ...] to choose a database connection. The list on the left shows the tables of the model, or all tables of the database when [All tables of the database] is checked.

- Enter a condition in [Filter (WHERE)] to see only some rows; [Clear filter] removes it.
- Click on a column title to sort the rows by this column.
- [Max. rows] limits the number of rows that are read.
- Double click on a value of a foreign key column to jump to the referenced row of the other table, following the relations of the model.

The Data Browser only reads; it does not change the database. The rows are copied into the grid, so no lock stays on the database while the window is open.

## Test Data Generator

The Test Data Generator fills the tables of the model with plausible rows: names, addresses, dates, numbers and texts that fit the names and datatypes of the columns. Foreign keys refer to existing rows, the lengths of the columns are kept.

- Select the tables and the number of rows per table ([Set for all] sets the same number for all tables).
- Choose the target database, the language of the data, the share of NULL values and a seed. The same seed gives the same script.
- Options: delete the existing rows first, leave auto increment columns to the database, COMMIT at the end.
- [Generate] writes the script of INSERT statements. It can be copied, saved, or executed in a database with [Execute in database ...]; the execution runs in one transaction and is rolled back on an error.

## Data Importer

The Data Importer imports text files into a table of a connected database: text with a separator (tabulator, semicolon, comma, space or another character, e.g. CSV files) or text in a fixed length format. On the Column Mapping page the fields of the file are assigned to the columns of the destination table; [Auto-Mapping] assigns fields with the same names. The settings can be stored as a preset and used again.

## HTML Report

The HTML Report writes a description of the model as an HTML page: the tables with their columns, and optionally the regions, the indices and the table options. The tables can be sorted by position or by name.

## Simple Web Front

The Simple Web Front creates PHP web pages for the tables of the model: lists of the rows, forms to enter and change them, with joins along the relations. The pages are grouped into views; the layout, the title and the output directory can be chosen. The database connection of the pages is entered on the first page of the plugin.

# Demo

The Demo plugin shows how a plugin reads the model; it writes the SQL create statements of the selected tables. It is meant as a starting point for developers who want to write their own plugin.

# Writing a Plugin

A plugin is a program named `DBDplugin_<Name>` (`DBDplugin_<Name>.exe` on Windows) in the directory of DBDesigner Fork. It is called with the file name of the model as its first parameter. The source code of the plugins in the directory `Plugins` of the project shows how the model classes of DBDesigner Fork are used.

# MCP Server for AI Assistants

The MCP server `DBDesignerMCP` lets an AI assistant such as Claude work with DBDesigner Fork models through the Model Context Protocol (MCP). The assistant can read and change models, write the SQL script, reverse engineer a database and compare a model with a database. The server uses the same model classes as DBDesigner Fork, so it reads and writes the model files and the SQL exactly like the program.

The server is a console program without a window. The AI assistant starts it and talks to it on its standard input and output; it is not started by hand.

## Installation

The MCP server is an optional part of the Windows setup program (task *Install the MCP server*, not preselected). It is installed next to the program as `DBDesignerMCP.exe`. Two more tasks register it in **Claude Code** (for the current user, all projects) and in **Claude Desktop**. The uninstaller removes the registrations again.

The registration can also be done later from a command prompt:

| | |
|---|---|
| `DBDesignerMCP --register-claude-code` | registers the server in Claude Code under the name `dbdesigner` (the `claude` command must be installed) |
| `DBDesignerMCP --register-claude-desktop` | adds the server to `claude_desktop_config.json` of Claude Desktop |
| `DBDesignerMCP --unregister-claude-code`<br>`DBDesignerMCP --unregister-claude-desktop` | remove the registrations |
| `DBDesignerMCP --version` | shows the version of the server, which is the version of the program |

Restart Claude Desktop after the registration. Other MCP clients can start `DBDesignerMCP.exe` without arguments as a local (stdio) server.

## Tools

The assistant first opens a model file (`open_model`) or starts a new one (`new_model`). All other tools work on this open model. Changes are kept in memory until `save_model` writes the model to its file or to another one.

| | |
|---|---|
| **Models** | open_model, new_model, save_model, get_model_settings, change_model_settings |
| **Tables and columns** | list_tables, describe_table, add_table, rename_table, change_table, delete_table, add_column, change_column, delete_column, get_standard_inserts, set_standard_inserts |
| **Indices and relations** | add_index, delete_index, list_relations, add_relation, delete_relation |
| **Diagram** | move_table, arrange_tables, list_regions, add_region, change_region, delete_region, list_notes, add_note, change_note, delete_note, list_images, add_image, change_image, delete_image, export_model_image (PNG, JPG or BMP) |
| **Datatypes** | list_datatypes, add_datatype, change_datatype, delete_datatype |
| **SQL** | export_sql: the SQL create script for a target database, with the options of the Export SQL Creates dialog |
| **Databases** | list_connections, connect_database, disconnect_database, list_database_tables, reverse_engineer, sync_database |

## Databases

The database tools work with MySQL, SQLite and Firebird. `list_connections` shows the database connections stored in DBDesigner Fork; a connection is best used by its name (its stored password is used). Otherwise the

driver, the database, the host, the port and the user are given directly; for SQLite and Firebird without host the database is a file. One connection is open at a time. The server reads the settings of the program but never changes them.

`sync_database` changes the database. Called without its `apply` parameter it only lists the statements it would execute, so the assistant (and you) can check them first. Tables that exist only in the database are kept.

Message boxes the program would show are not displayed by the server; their text is returned to the assistant as part of the result.

# Keyboard Shortcuts

## Navigation

| Key | Function | Info |
|---|---|---|
| F1 | Help | Calls the context sensitive Help. |
| Space | Panning | Holding down the Space-Key activates the panning function till the key is released. |
| Ctrl + Space | Zoom in | Holding down the Space-Key and the Ctrl-Key activates zoom-in function till the key is released. |
| Alt + Space | Zoom out | Holding down the Space-Key and the Alt-Key activates zoom-out function till the key is released. You can press Space + Ctrl + Alt, too. |
| Ctrl + Tab | Switch Workmode | Changes the Workmode from Design to Query and vice versa. |
| Ctrl + Shift + Tab | Switch Palettes | Press once to hide all palettes and again to show the palettes again. |
| Ctrl + Shift + 1..0 | Store Position Markers | Press Ctrl + Shift + 1,2, .. 0 to store a Position Marker. |
| Ctrl + 1..0 | Recall Position Markers | To recall a Position Marker press Ctrl + 1,2, .. 0. The position and zoom factor of the stored marker will be restored. |

# Design Mode

| Key | Function | Info |
|---|---|---|
| Ctrl + Q | Select Object(s) | Activates the Pointer cursor. In Design-Mode objects can also be moved with this cursor. |
| Ctrl + W | Move Object(s) | Activates the Move cursor. Needed to move objects in Query-Mode. |
| Ctrl + E | Scale Object | Activates the Scale cursor. Only used for Regions and Image objects. |
| Ctrl + T | New Table | Switches the mouse cursor to New Table Mode. |
| Ctrl + Shift + R | New 1:n Relation | Switches the mouse cursor to New 1:n Relation Mode. |
| Ctrl + R | New 1:n Relation (non identifying) | Switches the mouse cursor to New 1:n non-identifying Relation Mode. |
| Ctrl + Shift + Alt R | New 1:1 Relation | Switches the mouse cursor to New 1:1 Relation Mode. |
| Ctrl + A | Select All | Selects all objects |
| Ctrl + Del | Delete Selected | Deletes all selected objects. |
| Ctrl + Shift + C | Copy Highlighted Name | Copies the Highlighted Name to the Clipboard. |
| Ctrl + Z | Undo last Action | Undos the last Action |
| Ctrl + Shift + Z | Redo last Undo Action | Redoes the last Undo Action |
| Ctrl + O | Open model | Opens an existing model. |
| Ctrl + S | Save model | Saves the current model. The model can also be saved by clicking on the disk icon in the status bar. |
| Ctrl + Shift + S | Export SQL Script | Calls the SQL DDL Create Script Dialog to export the current model. |

# Query Mode

| Key | Function | Info |
|---|---|---|
| F9 | Execute SQL statement | Executes the current SQL statement. |
| Ctrl + Q | Select Object(s) | Activates the Pointer cursor. In Design-Mode objects can also be moved with this cursor. |
| Ctrl + S | Add Column to SELECT | Mouse Cursor to add Columns to the SELECT part of a SQL Command. |
| Ctrl + F | Add Column to FROM | Mouse Cursor to add Columns to the FROM part of a SQL Command. |
| Ctrl + W | Add Column to WHERE | Mouse Cursor to add Columns to the WHERE part of a SQL Command. |
| Ctrl + G | Add Column to GROUP BY | Mouse Cursor to add Columns to the GROUP BY part of a SQL Command. |
| Ctrl + H | Add Column to HAVING | Mouse Cursor to add Columns to the HAVING part of a SQL Command. |
| Ctrl + O | Add Column to ORDER BY | Mouse Cursor to add Columns to the ORDER BY part of a SQL Command. |
| Ctrl + E | Add Column to SET | Mouse Cursor to add Columns to the SET part of a SQL Command. |
| Ctrl + Shift + A | Copy SQL to Clipboard | Copies the current SQL Command to the Clipboard, formated for ASP. |
| Ctrl + Shift + D | Copy SQL to Clipboard | Copies the current SQL Command to the Clipboard, formated for Delphi/Kylix. |
| Ctrl + Shift + P | Copy SQL to Clipboard | Copies the current SQL Command to the Clipboard, formated for PHP. |

# DBDesigner Fork - Version History

The complete list of changes is in the README of the project and in the releases on GitHub.

| Version | Changes |
|---|---|
| next version | • Help menu: Visit GitHub Project opens the project page, Check for New Versions asks GitHub for the latest release.<br>• This documentation describes the current program. |
| 2.1.2<br>(Oct 2026) | • Help menu: Submit a bug / feature request opens the issues of the GitHub project. |
| 2.1.1<br>(Oct 2026) | • Reverse engineering and synchronisation keep much more of SQLite and Firebird: generated and computed columns, indices on an expression, descending and partial indices, CHECK constraints, STRICT, WITHOUT ROWID, COLLATE, ON CONFLICT, UNIQUE constraints, DEFERRABLE, the types of the databases, text defaults, global temporary tables.<br>• Example databases for SQLite and Firebird in the setup program.<br>• Bugfix: MCP server registration in Claude Desktop with an umlaut in the user name. |
| 2.1.0<br>(Oct 2026) | • New: MCP server for AI assistants, optional in the setup program.<br>• New: Reverse engineering creates a relation for a foreign key that refers to its own table.<br>• System requirement: Windows 10 or 11, 64 bit. |
| 2.0.2<br>(Oct 2026) | • New: Passwords of database connections can be stored (Windows).<br>• Dialogs can be resized and no longer lie above other programs.<br>• Bugfix: The HTML documentation opens with its pages again.<br>• Bugfix: Order of the tables in the model palette, error in the table editor on a column resize. |
| 2.0.1<br>(Oct 2026) | • New plugin: Data Browser.<br>• The page grid is visible across regions; the exported image has the size of the area of the objects.<br>• The window styles of CLX (Motif, SGI, Platinum) are hidden.<br>• Bugfix: Settings directory with an umlaut in the user name. |
| 2.0.0<br>(Oct 2026) | • First release of the port to Free Pascal / Lazarus, with a setup program for Windows (64 bit); Linux (GTK2) is built from the source code.<br>• Databases: MySQL 8, SQLite 3 and Firebird 3 or newer.<br>• High DPI support, German translation.<br>• New plugin: Test Data Generator.<br>• Setup program with the client libraries of SQLite, MySQL and Firebird (with the embedded engine). |
| 1.0 - 1.5<br>(2006 - 2010) | • DBDesigner Fork based on Delphi and Kylix, continuing DBDesigner 4. |

# DBDesigner 4 - Version History

## Source Releases

| Version | Changes |
|---|---|
| 4.0.5.5 | • Bugfix: Added option to deactivate quotes around default values.<br>• Bugfix: Added bug when fetching standard inserts from oracle.<br>• Bugfix: Fixed mouse cursor problem. |
| 4.0.5.4 | • Bugfix: The last placed table was not printed or displayed on the Navigator when placed on a region.<br>• Bugfix: A "List index out of bounds" error in the Relationship Editor.<br>• Bugfix: Enlarged Open/Save dialogs for Chinese and Korean Windows. |
| 4.0.5.3 | • Bugfix: Installer was missing midas.dll.<br>• Bugfix: Refresh was missing after drag'n'drop of datatype onto a column of a table on the canvas.<br>• New: Option to export Linux style linebreaks.<br>• New: Option to switch off datatype sync between FKs and PKs.<br>• New: When resizing the table editor the Comment column gets resized, too. |
| 4.0.5.2 | • Bugfix: Crow's Foot 1:1 Notation.<br>• Bugfix: Multibyte Language Reading Error.<br>• Bugfix: Region color is restored after discarding changes in the Region Editor.<br>• Bugfix: Improvment of the focus bug in Table Editor.<br>• New: Popup Menu entries in the Table Editor Column Grid to move columns up and down. |
| 4.0.5.1 | • Bugfix: Improvement of the focus bug in Table Editor.<br>• New: When using the a relationship tool the right mouse button can be used to choose a column as the Foreign Fey. This works only when there is only ONE Primary Key in the source table.<br>• New: Tables can be reordered in the Model Palette. New functions to sort tables by name and regions have been added.<br>• New: AutoIncrement is now turned on for Primary Keys by default. |
| 4.0.4.9 | • Bugfix: Using the Eraser tool caused AVs. fixed.<br>• Bugfix: Linux: Several libraries are now included with the distribution.<br>The following libraries have been added:<br>libpng.so.2.1.0.12<br>liblcms.so.1.0.9<br>libstdc++.so.5.0.0 |
| 4.0.4.8 | • Bugfix: Fixes the focus bug in Table Editor. After entering the last columns name the focus was to set back to the Column Grid. fixed.<br>• Bugfix: The Open/Save Dialogs should work on Win98 again. |
| 4.0.4.7 | • New: The Model Canvas Size can be edited in the Model Options Dialog.<br>• New: In Query Mode the Result Dataset can be printed to a PDF Document. |
| 4.0.4.6 | • Bugfix: When a column name or datatype is edited in the Table Editor and the Editor is closed before the change is submitted the new value is lost. fixed.<br>• Bugfix: Float values where not synchronised correctly. fixed.<br>• New: When deleting the selected Objects the names of the Objects are displayed before deletion.<br>• New: The Insert Key inserts a column in the Table Editor.<br>• New: The Splitter position in the Table Editor is now saved. |
| 4.0.4.5 | • Bugfix: When placing a model, the error message "Cannot focus a disabled or invisible window" was displayed. fixed.<br>• Bugfix: Tips were displayed in wrong language some time. Fixed.<br>• Bugfix: Some corrections of wrong assigned language texts. |
| 4.0.4.4 | • Bugfix: Setting path was not set so it was not able to change settings. fixed.<br>• Bugfix: Font issues finally addressed. fixed. |

| | |
|---|---|
| 4.0.4.3 | • Bugfix: Linux: When DBD4 was started the very first time, the directory ~/.DBDesigner4 was not present which caused DBD4 not to start. fixed.<br>• Bugfix: The Page & Printer Setup Dialog could not be displayed. fixed. |
| 4.0.4.2 | • New: Added new version of the SimpleWebFront Plugin 0.6.4.5<br>• New: Linux: linked against newer libqt.so.2.3.2 which enables smooth fonts on Linux.<br>• New: Export of SQL Optimize Tables Script and SQL Repair Tables Script.<br>• New: Several libraries have been removed from the Linux distribution because they are no longer needed.<br>• Bugfix: Table comments were not loaded from Model files. fixed.<br>• Bugfix: The Undo command after object(s) have been deleted was doing nothing. fixed.<br>• Bugfix: Modelfont was not saved. fixed. |
| 4.0.4.1 | • New: Objects can be linked from other Models and can be automatically refreshed.<br>• New: A new XML Parser is introduced to improve speed.<br>• New: Info text for Models stored in Databases.<br>• New: French translation by Alain Ségalou.<br>• Bugfix: Deleting a relation could lead to AV. fixed.<br>• Bugfix: After an update the existing DBConnections were not read and overwritten. fixed. |
| 4.0.3.29 | • Bugfix: Special Characters used in a Datatype's name or Description would cause a model not to open. |
| 4.0.3.28 | • Bugfix: Improved Cursor Key handling. Selected Objects were moved even when an editor was active. |
| 4.0.3.27 | • New: Cursor Keys move selected Objects. Shift+Ctrl move objects faster.<br>• New: 1:1 Non-Identifying added.<br>• New: Crow-Foot Notation.<br>• New: Region can be selected by click on Object Name.<br>• Bugfix: When a Table was connected to itself by a identifying Relation an unlimited number of recursive PKs were edit. Now only non-identifying Relations are allowed. |
| 4.0.3.26 | • New: Native MSSQL Reverse Engineering / Connectivity under Windows.<br>• New: Automatically collapse Tables while Reverse Engineering to save Canvas space.<br>• New: Improved Reverse Engineering Dialog.<br>• Bugfix: Oracle Relations missed PK - FK Mapping. fixed. |
| 4.0.3.25 | • Bugfix: When the language information was not set, the default was not set to english. fixed.<br>• Bugfix: When more than 10 files are opened these files cannot be added to the Recent Files Menu. fixed.<br>• Bugfix: The distribution did not contain the latest version of SWF. fixed. |
| 4.0.3.24 | • New: Added new version of the SimpleWebFront Plugin 0.6.2.<br>• New: Close All menu item in the file menu.<br>• Bugfix: Several Bugfixes in the Database Synchronation function.<br>• Bugfix: Options in the Reverse Engineering dialog now disabled when not selected. |
| 4.0.3.23 | • Bugfix: Quotation Character was not used when reverse engineering a MySQL database. fixed.<br>• Bugfix: When storing a model in a database running on Linux, the ' characters were removed. fixed.<br>• Bugfix: Double-clicking in the DB Model Palette was only working with tables. fixed. |
| 4.0.3.22 | • Bugfix: Reference Definition for InnoDB Tables were not created because of different syntax. This can now be adjusted in the Options Dialog.<br>• New: Application Font can be changed.<br>• New: Tables can be collapsed.<br>• New: Shortcut for exporting SQL Script (CTRL+SHIFT+S).<br>• New: When exporting a SQL Script, Regions can be selected. |
| 4.0.3.21 | • New: Message asking the user to save the model after a plugin was called.<br>• New: New Menu Entry in the Column Popup Menu in the Table Editor to add column(s) to the current index.<br>• New: Improved table shadows under Linux. |

• Bugfix: Drag'n'Drop of columns onto the index column list in the Table Editor may cause an Index out of Bounds error on some machines. fixed.

4.0.3.20
• Bugfix: When using the DBSync function all columns were modified. fixed.
• Bugfix: Improved DataImporter DBConn handling.

4.0.3.19
• Bugfix: Fixed filelist in the Setup.ini.

4.0.3.18
• New: Physical Column Mapping to enable a e.g. Datatype named GUID which is automatically changed to BIGINT when exported/synchronized.
• New: Default Datatype substitution for ODBC Databases in the Reverse Engineering Dialog.
• New: Option to disable Model-Change state after a SQL Command is executed and a SQL History Entry is made.
• Bugfix: There was not space between tablename and ( in the SQL Create/Insert command causing SQLErrors when using function names as tablenames. fixed.
• Bugfix: Hostname / Hostcaption mixup fixed.
• Bugfix: Reverse Engineering an Oracle Database causes an AV when there is more then one relation in the database. Fixed.
• New: Icon for Geographic Datatypes.

4.0.3.17
• New: Added Column Prefix / Postfix functions to the Table Column's Popup Menu in the Table Editor.
• New: When several Table Columns are selected in the Table Editor and a datatype with no parameters required is dragged onto the Column Grid, the datatype is assigned to all selected columns.
• Bugfix: When printing, column icons were painted at the wrong place if Show Foreign Keys was turned off. fixed.
• Bugfix: Using a real n:m relation may lead to AV when automatic index generation for FK fields is turned on. fixed.
• Bugfix: When selecting several Table Columns in the Table Editor and pressing the right mousebutton, only one Column was selected. fixed.
• Bugfix: When removing a Relation, the Foreign Key was not removed from user defined indices causing AVs when exporting SQL Scripts. fixed.

4.0.3.16
• New: SQL Syntax Highlighting.
• New: Second SQL Query Layout.
• New: 8 Temporary SQLCommand Drop Targets.
• Bugfix: Impropper DateTime values may lead to infinite error message loop. Improved.
• Bugfix: Linux: When closing the program unassigned ActiveControl may lead to AV. fixed.

4.0.3.15
• New: Scaling of the Index tab in the Table editor when the whole editor is resized.
• New: Oracle 8 compatibility when reverse engineering a Oracle database.
• Bugfix: Recursive relations on a table itself caused various AVs and strange behaviour. fixed.
• Bugfix: When DBDesigner4 was launched the very first time the DBDesigner4_Translations.txt wasn't copied to the personal files directory. This caused a warning. fixed.

4.0.3.14
• New: Translation features were reprogrammed and enabled.
• Bugfix: Infinite Loop could cause a Stack Overflow error when two tables were connected by a relation and were moved to a special position to each other.

4.0.3.13
• New: Geographic DataTypes added.
• New: When a object is copied to / pasted from the clipboard, the Status Bar displays a corresponding message.
• Bugfix: Deleting a table in the Tables Tab did not remove the table from the list. That may lead to an AV. fixed
• Bugfix: Copying object(s), Saveing the model in the database caused the model's save path to be changed. fixed.
• Bugfix: QApplication_postEvent caused AVs because of improper use. fixed.
• Bugfix: When a model was opened from database the stored model was added to the current model. fixed.

4.0.3.12
• New: Oracle Reverse Engineering functions.
• New: Recursive Relations to enable a relation between a table and itself.
• Bugfix: Translation.ini was bigger than 64kB causing errors when using Win98.

| | |
|---|---|
| 4.0.3.11 | • New: When entering new table columns, the cursor jumps between column name / datatype automatically.<br>• New: Copy selected Objects as Image function copies the image to clipboard for use in Office / Photoshop / ...<br>• New: Export selected Objects as Image function.<br>• New: Export Model as Image now only exports the model area, not the complete model canvas. |
| 4.0.3.10 | • Bugfix: When exporting grid-data as SQL INSERTS the tablename could contain the whole SQL Select command. fixed. |
| 4.0.3.9 | • Bugfix: Disabled Translation function because they might cause Access Violations. |
| 4.0.3.8 | • Bugfix: The default table prefix was added to the tablename, causing broken DBSync / Export functions. fixed. |
| 4.0.3.7 | • New: Schema Filter in the Reverse Engineer Dialog.<br>• Bugfix: When a table prefix was deleted and previously assigned to a table, exporting SQL Scripts causes an Index out of Bounds error. fixed. |
| 4.0.3.6 | • Bugfix: When using a Linux MySQL Server one could not list a host's databases or create a new one. fixed.<br>• Bugfix: Linux: Column Grid in the Table Editor missed Captions. fixed. |
| 4.0.3.5 | • New: ESC Key in Editors closes the editor without applying changes.<br>• New: Table Editor now supports improved Datatype editing.<br>• Bugfix: When a Table is renamed the FK Columns were not updated. fixed. |
| 4.0.3.4 | • New: ERwin 4.1 XML import<br>• New: Constraint Object move when user holds the Shift Key.<br>• New: Center Model function in the Edit Main Menu.<br>• New: New Option to select a different HTML Browser for the HTML Documentation.<br>• New: New Option to enable "back ticks" for table/column/index names<br>• New: New Option to disable reserved words replacement<br>• New: Localisation<br>• Bugfix: DBSyncronize raised an AV when a table with a very long name was synchronized. fixed.<br>• Bugfix: DBSyncronize now treats tinyint(1) as BOOL.<br>• Bugfix: FK Datatype and Options were not updated when the source Key was changed. fixed.<br>• Bugfix: Win98/ME: Open File Dialog raised exception: Drive C: not valid. |
| 4.0.3.3 | • Bugfix: Compare/Alter Statisic after Database Synchronisation listing was incorrect. fixed.<br>• Bugfix: Scripts are not executed correctly when using ; as seperator. fixed. |
| 4.0.3.2 | • New: New option [Minimize on [Ctrl+Shift+C] Copy Command].<br>• New: Compare/Alter Statisic after Database Synchronisation.<br>• Bugfix: AV when closing About Window. |
| 4.0.3.1 | • New: When executing a query the affected rows and the execution time is displayed.<br>• New: SQL Scripts can now be executed.<br>• New: Reserved words can now be edited in the DBDesigner4 Options dialog.<br>• New: Automatic Index creation on FK Columns.<br>• New: Foreign Key Columns are locked in the table editor.<br>• New: Table Editor can be resized now.<br>• New: New design of the Table Options / DBDesigner4 Options.<br>• New (Windows): The DBDesigner4 Settings directory is now located in the user's Application Data directory. Existing settings are moved from the users Personal Files directory automatically.<br>• New: OK / Cancel Buttons in Editors and Option Dialoges.<br>• Bugfix: Exception when syncronizing Standard Inserts and they were not seperated by a clear line.<br>• Bugfix: When assigning a datatype to a column in the table editor, an additional click on the datatype palette was required before a datatype could be dragged. fixed. • Bugfix: When entering several new columns in the table editor, the only way to quit editing was to press ESC. Now Return in an empty field will chancel the edit mode. • Bugfix: When an object was deleted when opened in an editor, an Access Violation occured. fixed. |

| | |
|---|---|
| | • Bugfix: Relation Kind was not restored when undo was selected. fixed.<br>• Bugfix: Plugin Parameters where not displayed entirely in the Model Options. fixed. |
| 4.0.2.96 | • New: Ctrl+Shift+Tab now works when Palettes are docked.<br>• New: Automatic fix of wrong ENUM/SET definition (see 4.0.2.95 release info)<br>• Bugfix: Datatype's parameters could not be editied on Linux machines<br>• Bugfix: Foreign Key Reference no longer contains MATCH FULL/PARTIAL option<br>• Bugfix: ENUM datatype's parameters could be truncated in table editor |
| 4.0.2.95 | • New: Added ForeignKey Prefix / Postfix options.<br>• New: Added support for ENUM / SET datatypes.<br>• Bugfix: Foreign Key Reference Syntax now works with MySQL 4.0<br>• Bugfix: Foreign Key Reference is now created when using DBSync function. |
| 4.0.2.94 | • New: Added Default Table Type and Default ActivateRefDefForNewRelation options. |
| 4.0.2.93 | • New: Make DBDesigner4 work with no personal settings for each user with the -disablePersonalSettings parameter at startup. |
| 4.0.2.92 | • New: Use Table Name In Foreign Key Columns option.<br>• New: The Default Values for Snap to Grid and TableNameInRefs can be modified. |
| 4.0.2.91 | • Bugfix: When a model was saved after Reverse Engineering an access violation occured. Fixed. |
| 4.0.2.90 | • New: Snap to Grid feature. |
| 4.0.2.89 | • Bugfix: IDs of Objects could overlap after model was loaded. Fixed. |
| 4.0.2.88 | • Bugfix: Plugin - Data was not loaded. Fixed.<br>• New: When using the Database Synchronisation function tables are now renamed if the name was changed in the model. Previously they had beed dropped and recreated.<br>• New: Index Columns now support the additional lenght parameter for BLOB fields. |
| 4.0.2.87 | • Bugfix: Version number was not automatically increased when model was stored in database and option was selected. Fixed.<br>• New: The model's version can now be edited in model option dialog. |
| 4.0.2.86 | • First puplic release. Version history will start from this version. |

# Documentation Releases

| **Version** | **Changes** |
|---|---|
| 1.0.42 | • Renamed syncrosation to synchrosation.<br>• Added Plugins Menu |
| 1.0.41 | • New Modelling and Quering Section. |
| 1.0.40 | • Updated Screenshots and listing of new functions. |
| 1.0.39 | • Improved Fundamentals section. Included GNU GPL with the Online and PDF documentation. |
| 1.0.38 | • First puplic release. Version history will start from this version. |

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Version 2, June 1991

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